

6

Wave Nature of Light and Electromagnetic Waves

In this chapter, we will learn about

- the wave nature of light
- the interference patterns formed by a double slit and a plane transmission grating
- the properties of electromagnetic waves

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6.1 Wave nature of light

Lasers are useful when studying the nature of light. It is a concentrated and monochromatic beam of light (Fig 6.1a). Since laser beam is very energetic, exposure to it may cause serious damage to our eye and skin. Therefore, you must be very careful when using lasers.

Make sure you follow all the safety precautions below when using lasers.

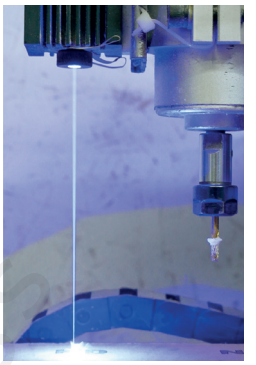


Fig 6.1a *Laser beam is emitted.*



Safety precautions

- Do **not** view a laser beam through a single slit / double slit / grating!
- Do **not** view the source of a laser beam directly and no part of the skin should be exposed to the direct beam.
- All persons who may enter or work in the laser area and are liable to be exposed to laser radiation should wear appropriate protective goggles.
- The laser beam path should be positioned well above or below the eye level.
- The Radiant power of the lasers should not exceed 1 mW.
- Protective screens should be used to prevent teachers and students from being exposed to the reflection of the laser beam.
- When a laser is in operation, warning signs should be displayed in conspicuous locations both inside and outside the experiment area and on doors giving access to the area.

✎ Ts are suggested to give a brief review about diffraction before demonstrating the diffraction of light.

1 Diffraction of light

✎ The central bright fringe in a diffraction pattern is wider than the subsidiary fringes. This is different from an interference pattern which has fringes of equal width.

DSE exam

16(1A)Q19, 18(1A)Q16

Flipped classroom

Watch a video and answer the questions.

There are dark and bright bands in the diffraction pattern. The reason for their formation is beyond the scope of this book.

Light undergoes diffraction like water waves. It spreads out after passing through a narrow gap (Fig 6.1b). Let us study the diffraction of light in this section.

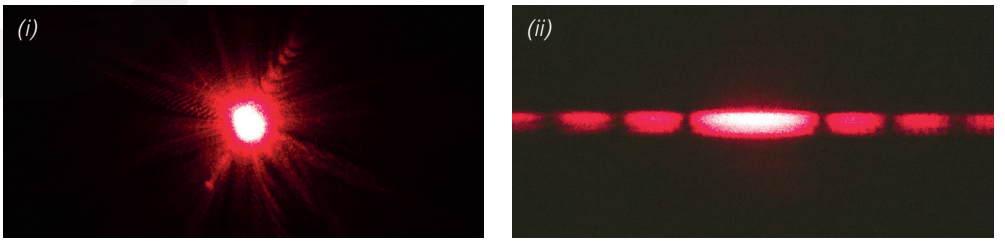


Fig 6.1b *The light pattern formed on the wall when a laser beam (i) is directly shone on the wall, and (ii) passes through a narrow gap.*

Experiment 6a / Factors affecting the degree of diffraction of light

1 Direct a laser beam onto a single slit (Fig a). Capture the pattern on a far screen (Fig b).

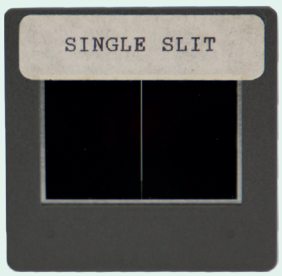


Fig a



Fig b

2 Repeat with single slits of different widths. Observe how the width of the central bright spot changes.

3 Repeat with lasers of different colours. Observe how the width of the central bright spot changes.

4 Alternatively, replace the laser with a ray box. View the filament through single slits of different slit widths.

Discussion

How does the diffraction pattern change when (a) the slit width changes, (b) the colour of the laser beam changes?

Video

→ This video demonstrates Expt 6a.

You must be very careful when using laser in experiments. Make sure you follow all the safety guidelines on p.102.

a Effects of slit width

In Chapter 5, we learned that the degree of diffraction depends on the width of the gap. When the gap is narrower, the wave spreads more (Fig 6.1c). The diffraction of light behaves in the same way (Fig 6.1d).

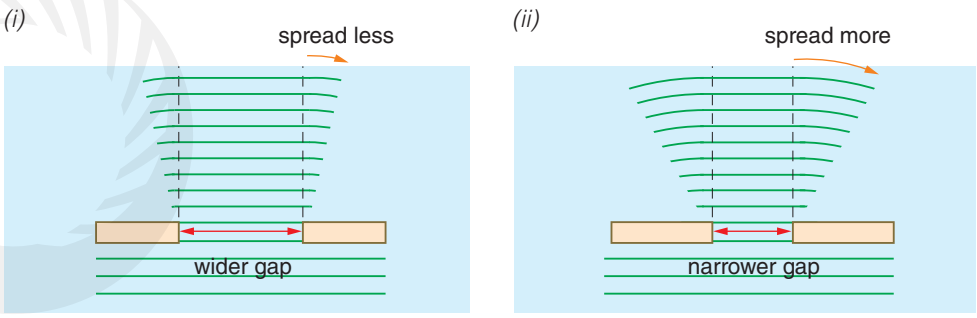


Fig 6.1c *Diffraction of waves by (i) a wider gap and (ii) a narrower gap.*

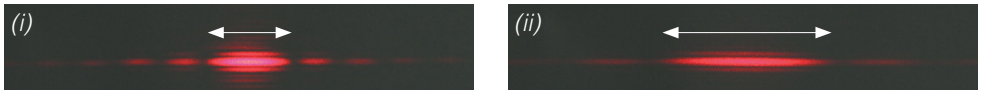


Fig 6.1d *Diffraction of light by (i) a wider slit and (ii) a narrower slit.*

Simulation

→ This simulation demonstrates the diffraction of different coloured lights. It is available on OUP web in PC version (.exe) only.

b Effects of colour

The degree of diffraction also depends on the wavelength of the wave. When the wavelength is longer, the wave spreads more (Fig 6.1e).

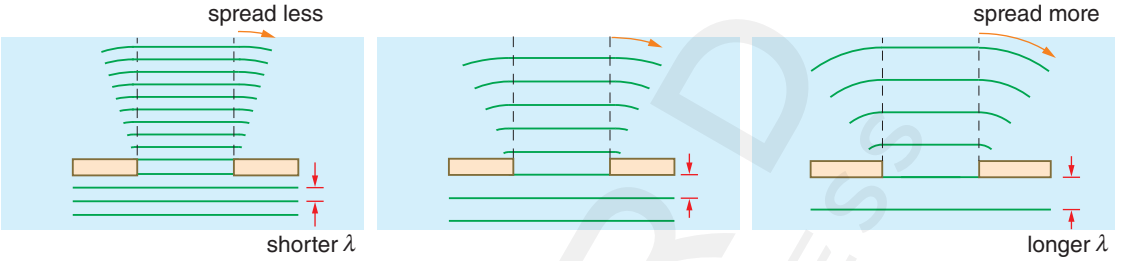


Fig 6.1e *Wavelength affecting the degree of diffraction.*

Figure 6.1f shows the diffraction pattern produced when light of different colours passes through the same slit. Blue light spreads out the least and red light the most whilst green light spreads out at an angle in between the other two. This suggests that when light is considered as wave, the wavelength increases from blue to green to red.



Fig 6.1f *Diffraction patterns obtained by (i) blue, (ii) green and (iii) red light.*

✎ Ts are suggested to give a brief review about interference before demonstrating the interference of light.

2 Interference of light

When a laser beam passes through two narrow slits very close to each other, which is called a **double slit**, bright and dark **fringes** (bands of colour) are formed (Fig 6.1g). This pattern can be explained by considering light as a wave: Light waves are diffracted at the two slits and interference occurs. The bright fringes correspond to constructive interference and the dark fringes corresponds to destructive interference. We will study the interference of light in details in Chapter 6.2.

✎ Since the intensity of the fringes drops to a minimum and increases again due to the effect of diffraction, many students observing from a distance may miss the fine detail of the interference pattern and consider the envelopes of the diffraction pattern as the interference pattern.

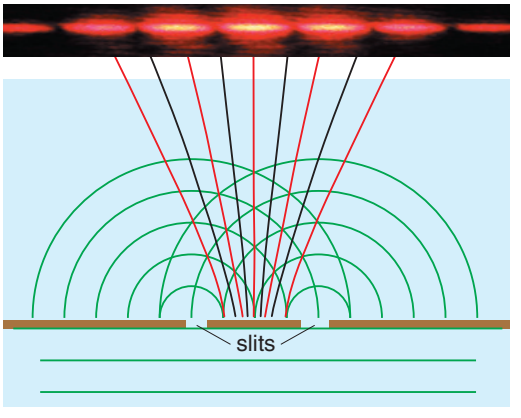


Fig 6.1g *Interference pattern of light explained.*

→ This is a Video Quiz about the experiment 'Interference of light'.

double slit 雙縫　　fringe 條紋

Quick check 1

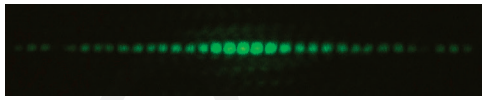
→ E-version in OUP Exercise Platform is also available.

Google Forms | Microsoft Forms

1 Green light is directed on a film with a slit and a diffraction pattern is obtained on a screen. Which of the following changes will result in a wider pattern? Put a tick in the corresponding box(es).

- ☐ Replace the film with another one with a narrower slit.
- ☐ Replace the light source with a blue one.
- ☐ Put the light source further away from the film.
- ☐ Put the screen further away from the film.

2 When a green laser beam passes through a double slit, the following pattern in is formed.



Determine whether the following statements are true.

(a) Diffraction of light occurs. (T/F)

(b) Interference of light occurs. (T/F)

(c) The dark fringes are caused by the destructive interference of light. (T/F)

DSE goal

Point out light as an example of transverse wave.

State the range of wavelengths for visible light.

Realize diffraction and interference as evidence for the wave nature of light.

Cross-link

We will study electromagnetic waves in Ch 6.4.

3 Light as a wave

Light does exhibit diffraction and interference. This provides evidence that light is a wave. In fact, light is an **electromagnetic wave**, which is a **transverse wave**. Further studies show that the wavelength of light lies in the range from about **400 nm** (**violet** end) to **700 nm** (**red** end) (Fig 6.1h).

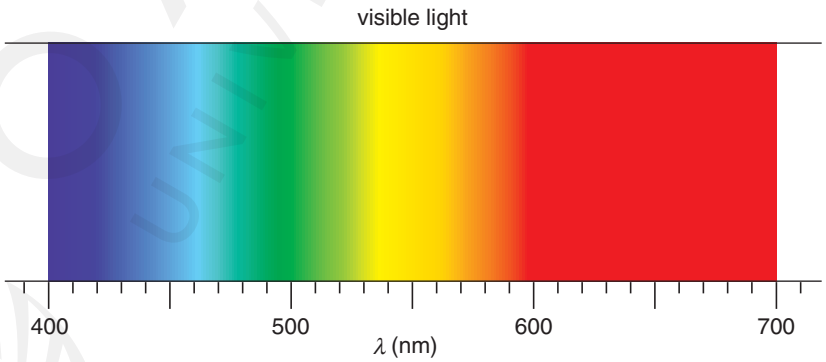
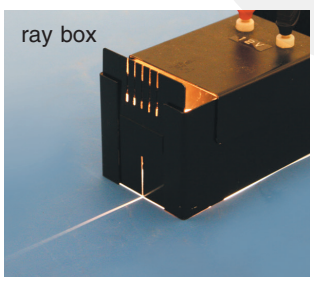


Fig 6.1h *Wavelength of light.*

The light ray from a ray box does not seem to diffract after passing through a slit because the wavelength of light is much shorter than the slit width.



Recall that the diffraction of water waves is observable only when the width of the gap is comparable to the wavelength of the waves. If the wavelength is much shorter than the width of the gap, the wave will pass through the gap with almost no diffraction.

Light waves behave similarly. Since the wavelength of light (~10^{-7} m) is much shorter than the gaps we encounter in daily life, we seldom observe the diffraction of light. In Experiment 6a (p.103), the slits are very narrow (~0.1 mm), so the diffraction of light can be observed.

electromagnetic wave 電磁波

We use arrowed lines perpendicular to wavefronts to indicate the direction of motion of a water wave. We do the same for a light wave (Fig 6.1i). The arrowed lines are the light rays that we learned about in Book 3A .

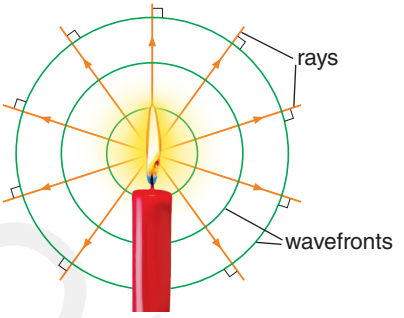
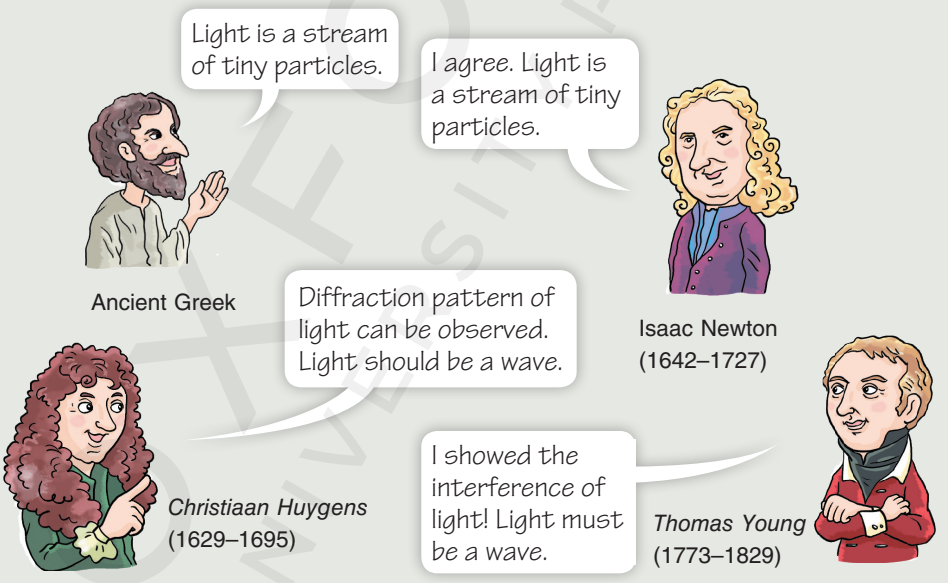


Fig 6.1i *Light can be represented by wavefronts or by rays.*

Nature of science

Collision of great minds

There was a huge debate among scientists on the nature of light.

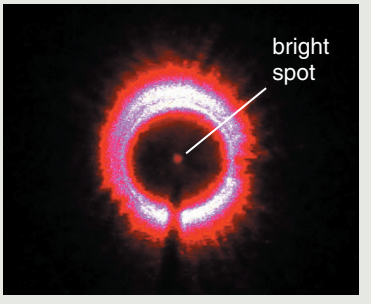


Watch this video about this effect. ▶

Video

→ This video demonstrates a bright spot appeared at the centre of a circular object's shadow.

Later, *François Arago* (1786–1853) showed in an experiment that a bright spot appeared at the centre of a circular object's shadow. Since then, scientists gradually accepted that light is a wave.



Cross-link

We shall learn more about this in Book E2 Ch 1. ▶

However, the story did not end here. *Albert Einstein* (1879–1955) discovered that light also shows the properties of particles.

Now we believe that light can behave as both particles and waves.

This journey of scientific discovery tells us that:

- It is evidence that matters, not the status of a scientist.
- Scientific knowledge is ever changing, sometimes in rather revolutionary ways.
- Scientific knowledge is hard work built upon the hard work of others.

Christiaan Huygens 惠更斯　Thomas Young 楊格　François Arago 阿拉戈
Albert Einstein 愛因斯坦

Google Forms Microsoft Forms

Quick check 2

→ E-version in OUP Exercise Platform is also available.

1 Which of the following support(s) the notion that light is a wave? Put a tick in the corresponding box(es).

- ☐ When light is reflected, the angle of reflection equals the angle of incidence.
- ☐ When light passes a narrow slit, it spreads into the shadow of the slit.
- ☐ When a laser beam passes through a double slit, bright and dark fringes are formed.

Exercise 6.1

1 Why is the diffraction of light difficult to observe in everyday life?

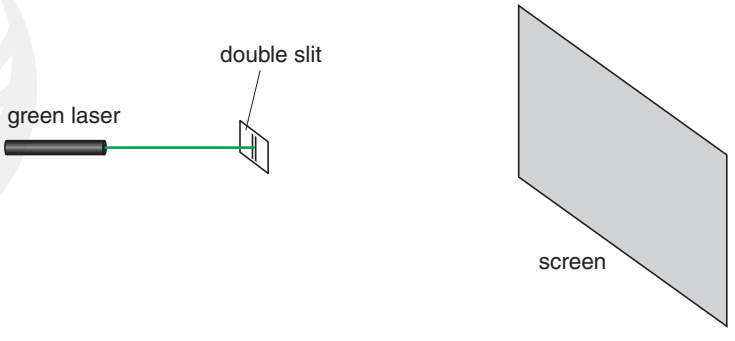
A The speed of light is very high.
B The wavelength of a light wave is very short.
C Only the light emitted by a laser shows diffraction.
D All of the above

★**2** Which of the following statements is/are correct?

(1) Light is a transverse wave.
(2) The wavelength of red light is about 7×10^{-7} m.
(3) Violet light has a longer wavelength than red light.

A (1) only
B (3) only
C (1) and (2) only
D (2) and (3) only

★**3** A green laser beam is directed onto a double slit as shown. A pattern is observed on the screen behind the double slit.



(a) Describe the pattern observed on the screen.
(b) Explain how the pattern is formed.

6.2 Young's double slit experiment

In chapter 6.1, we made a double slit and showed the interference of light. This is like the famous experiment conducted by Thomas Young in 1801. His experiment played a major role in establishing the wave nature of light, and is now called **Young's double slit experiment** (Fig 6.2a).

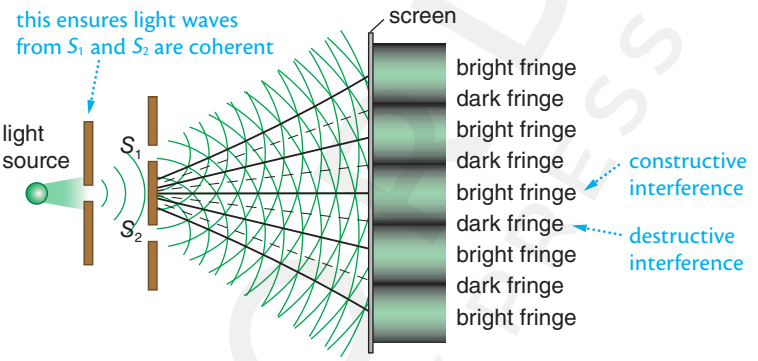


Fig 6.2a *Young's double slit experiment (top view, not to scale).*

DSE goal

Examine the interference patterns in the Young's double slit experiment.

DSE exam

20(1A)Q15

Light waves from the two slits are always coherent if a laser beam is used. ▶

Video

→ This video demonstrates Expt 6b.

You must be very careful when using laser in experiments. Make sure you follow all the safety guidelines on p.102.

1 Examining the pattern

Experiment 6b / Interference patterns formed by double slits

1 Direct a laser beam onto a double slit (Fig a). Capture the pattern on a screen (Fig b).

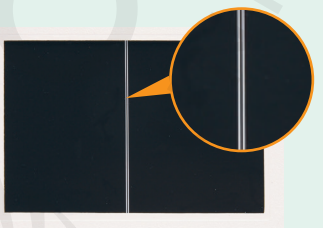


Fig a

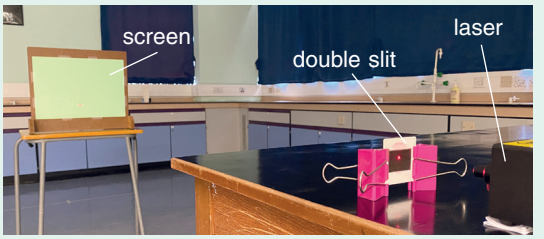


Fig b

2 Repeat with double slits of different slit separations.

3 Repeat with lasers of different colours.

4 Repeat by putting the screen at different distances from the double slit.

5 Alternatively, replace the laser with a ray box. View the filament through double slits of different slit separations.

Discussion

How does the interference pattern change when (a) the slit separation, (b) the colour of the laser beam and (c) the distance between the double slit and the screen is changed?

Young's double slit experiment 楊氏雙縫實驗

Figure 6.2b shows the interference pattern produced by a double slit.

1 It contains a number of bright fringes.

2 The pattern is symmetric. There is always a bright fringe at the centre of the pattern. This fringe is called the central bright fringe.

3 The fringes are **evenly spaced**, i.e. the fringe separation Δy is the same across the whole pattern.

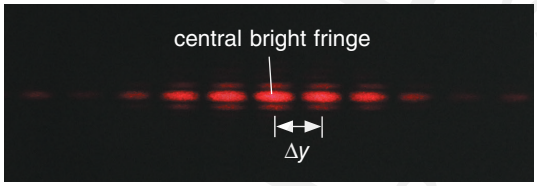


Fig 6.2b *Interference pattern produced by a double slit.*

For a typical double slit in the laboratory,
slit width $\leq 10^{-4}$ m
slit separation $\leq 10^{-3}$ m
If the slit separation is too large, the fringes will be so close that they become indistinguishable.

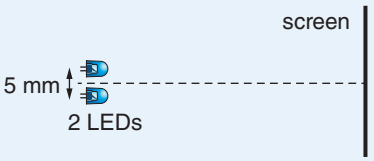
▶ To produce an interference pattern, the slit width must be comparable to the wavelength and the two slits cannot be too far apart. To make the pattern **observable**, the light waves passing through the two slits must be **coherent** and have **similar amplitudes**.

The light waves from two light sources may have the same frequency but the phase difference is not constant. The reason for this is out of the scope of this book.

▶ Note that the experiment **cannot** be done by replacing the double slit with two light sources (e.g. two LEDs). Light waves from **two light sources** are **not coherent** and the interference pattern is **not observable**.

Right and wrong

Two LEDs emitting light waves of the same frequency are 5 mm apart.



✗ An interference pattern can be observed on the screen.

✗ The light waves do not interfere with each other.

✓ The light waves do interfere with each other. However, the interference pattern changes so fast that it is not observable. To the human eye, the screen in front of the LEDs is uniformly illuminated.

Know more

Thomas Young

Young was a British scientist. He made important contributions in different fields. He was credited with establishing the wave theory of light.



DSE exam
21(1A)Q16

Simulation
→ This simulation shows the interference pattern formed by a double slit. Ss can choose different colours of light and vary the slit separation to see the effect on the interference pattern. It is available on OUP web in PC version (.exe) only.

2 Factors affecting the pattern

a Slit separation

In Experiment 6b, we can see that when the slit separation a decreases, the fringe separation Δy increases (Fig 6.2c). This is similar to the interference of water waves (Fig 6.2d).

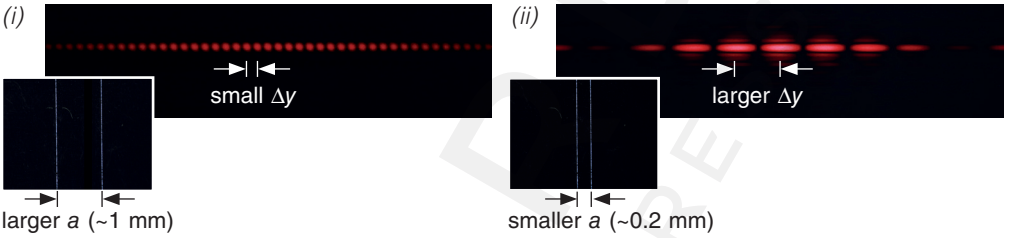


Fig 6.2c *Interference patterns produced by a double slit with (i) a larger slit separation and (ii) a smaller slit separation.*

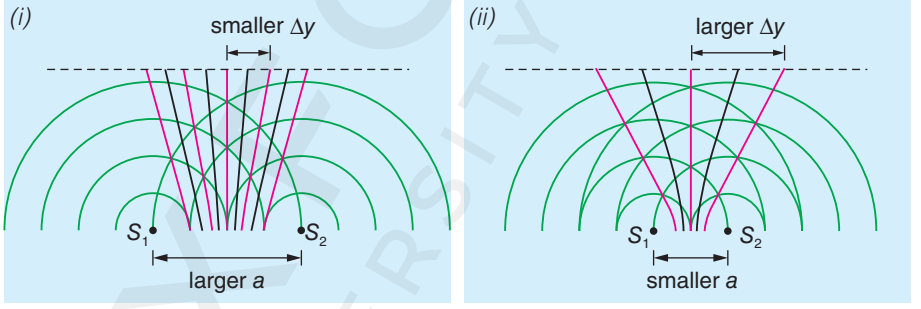


Fig 6.2d *Interference patterns of water waves when the separation between the sources is (i) larger and (ii) smaller.*

b Wavelength

When the wavelength λ of light increases, Δy increases (Fig 6.2e). You can see similar changes in the interference of water waves (Fig 6.2f).

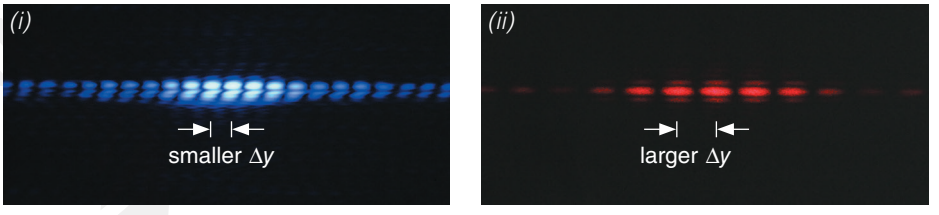


Fig 6.2e *Interference patterns of (i) blue light (shorter λ) and (ii) red light (longer λ).*

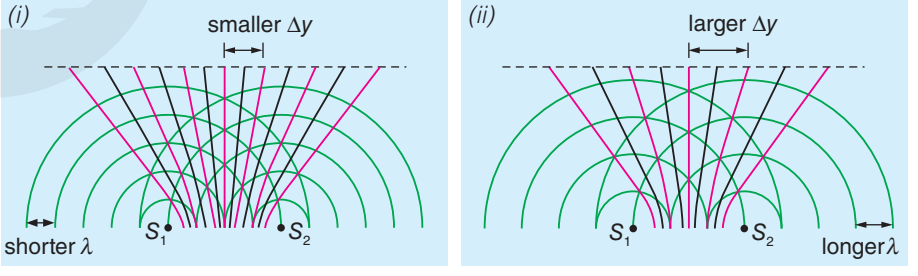


Fig 6.2f *Interference patterns of water waves with (i) a shorter λ and (ii) a longer λ.*

C Slit-screen distance

When the distance D between the double slit and the screen increases, Δy increases (Fig 6.2g). Similar effect can be observed in the interference pattern of water waves (Fig 6.2h).

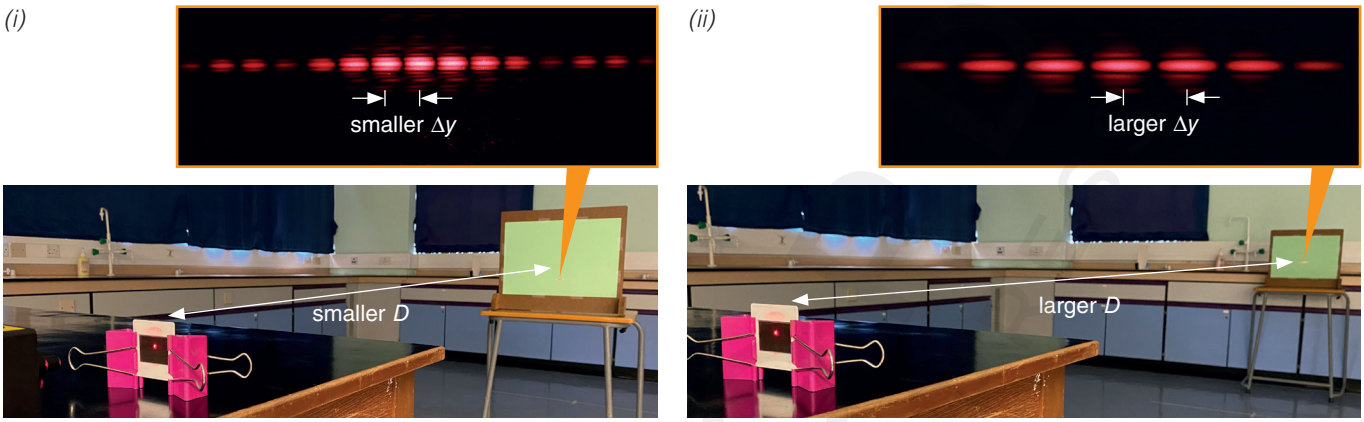


Fig 6.2g Interference patterns of light formed on screens with (i) a smaller D and (ii) a larger D.

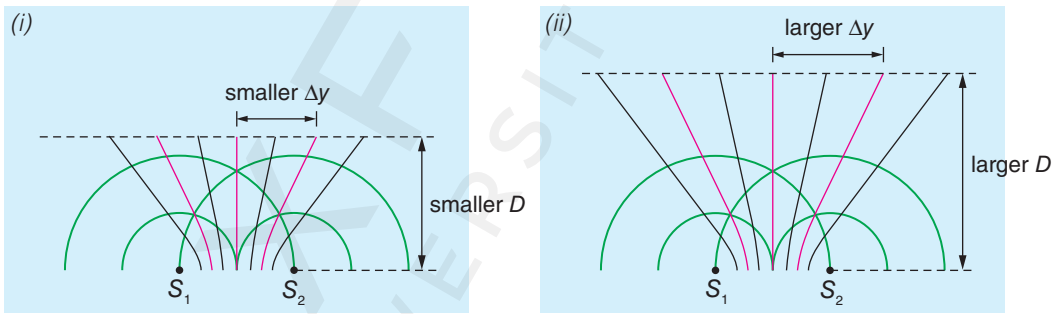


Fig 6.2h Interference patterns of water waves with (i) a smaller D and (ii) a larger D.

In the interference pattern formed by a double slit, the fringe separation Δy increases when:

1 the slit separation a decreases;

2 the wavelength λ of light increases;

3 the distance D between the double slit and the screen increases.

Quick check 3

→ E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

1 A student directs a green laser beam towards a double slit to produce an interference pattern on a screen. What can he do to increase fringe separation? Put a tick in the corresponding box(es).

- ☐ Use a double slit with a larger slit separation.
- ☐ Use a double slit with a larger slit width.
- ☐ Move the screen further away from the double slit.
- ☐ Move the laser pointer further away from the double slit.
- ☐ Use a blue laser instead of a green laser.

Ext

DSE goal

Apply $\Delta y = \frac{\lambda D}{a}$ to solve problems.

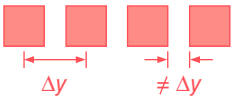
DSE exam

15(1B)Q7, 16(1B)Q6(a), 23(1A)Q17

→ Start Oxford Physics XR 'Double slit and grating' and scan this page to play.

iOS Android

Remind Ss that the fringe separation (and slit separation) is the distance between the centres of two fringes (slits), not the width of the dark region between them.



3 The fringe separation

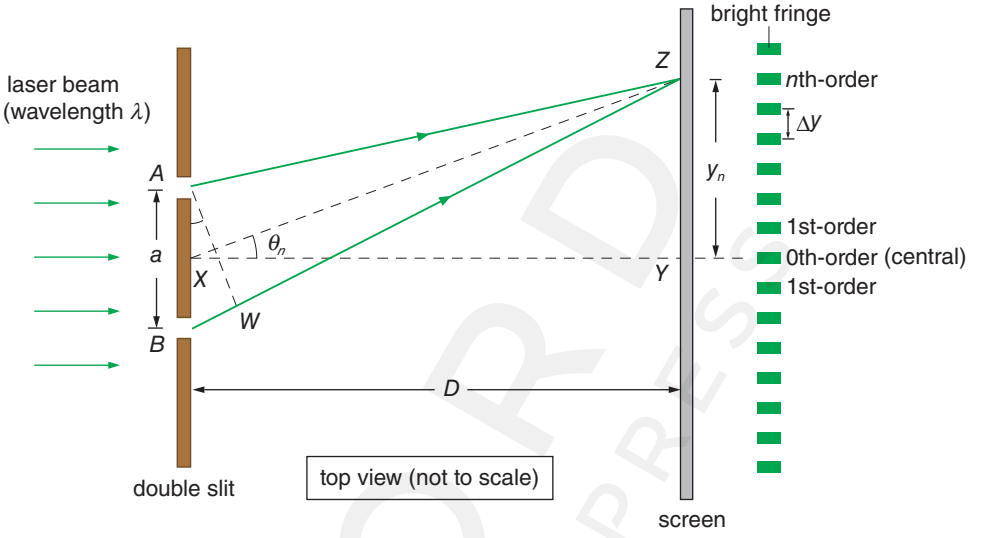


Fig 6.2i *Investigating the relationship between Δy, λ, D and a*

Consider a monochromatic light (i.e. light of a single wavelength, such as a laser beam) of wavelength λ being directed towards a double slit of slit separation a (Fig 6.2i). A screen is placed at a distance D from the double slit and an interference pattern is formed on the screen. The fringe separation Δy can be expressed as:

$$\Delta y = \frac{\lambda D}{a}$$

(λ: wavelength (m); D: distance of screen from double slit (m); Δy: fringe separation (m); a: slit separation (m))

Δy obtained from this formula also equals the distance between adjacent dark fringes. ▶

This formula is applicable, and Δy remains the same for the entire pattern, only if **both** conditions below are satisfied:

1 D is much greater than a ($D >> a$).

2 θ_n is very small, i.e. the size of the whole interference pattern on the screen is much smaller than D ($y_n << D$).

This agrees with the results of Experiment 6b (p.108–109). ▶

This formula suggests that Δy increases when

1 a decreases;

2 λ increases;

3 D increases.

We usually number the central bright fringe as the 0th-order bright fringe, the two fringes next to it as the 1st-order bright fringes, and so on. Moreover, when $D >> a$, for the nth order bright fringe,

$$a \sin \theta_n = n\lambda$$

Know more

Deriving $\Delta y = \frac{\lambda D}{a}$

We can derive this formula.

▶ p.149

monochromatic light 單色光

Ext

Video

→ This video shows how to estimate the wavelength of light using a double slit.

→ This is a Video Quiz about the experiment 'Estimating the wavelength of light using a double slit'.

Example 1 Estimating the wavelength

A red laser and a blue laser are directed towards a double slit of slit separation 0.3 mm in turn. The figures below show the interference patterns produced on a screen 6 m from the double slit.

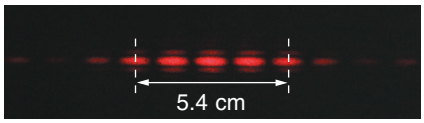


Fig a

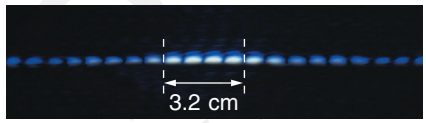


Fig b

(a) State the advantage of measuring the separation between several fringes instead of two adjacent fringes in estimating the wavelength of the light.

(b) Estimate the fringe separation in each case.

(c) Estimate the wavelengths of the red light and the blue light.

Solution

(a) This reduces the percentage error in the measurement.

(b) For the red laser, fringe separation $= \frac{5.4}{4} = 1.35$ cm

For the blue laser, fringe separation $= \frac{3.2}{4} = 0.8$ cm

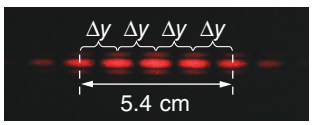
(c) By $\Delta y = \frac{\lambda D}{a}$, $\lambda = \frac{a\Delta y}{D}$.

Wavelength of red laser $= \frac{(0.3 \times 10^{-3})(1.35 \times 10^{-2})}{6} = 6.75 \times 10^{-7}$ m

Wavelength of blue laser $= \frac{(0.3 \times 10^{-3})(0.8 \times 10^{-2})}{6} = 4 \times 10^{-7}$ m

Cross-link
You learned percentage error in Book 2 Ch 1.

Tip
$4(\Delta y) = 5.4$ cm



We can estimate λ by directing the light towards a double slit with known value of a and measuring Δy and D.

Caution
Be aware of the units.

Quick check 4

Google Forms Microsoft Forms

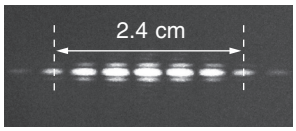
1 Complete the following table.

Δy / cm	λ / nm	D / m	a / mm
0.6		1.5	0.1
0.4	450	2	
	632	6	0.2

2 *True or false:* The formula $\Delta y = \frac{\lambda D}{a}$ is applicable when a is much smaller than D. (T/F)

Learn by Practice 1

1 A laser of wavelength 400 nm is directed towards a double slit. An interference pattern is formed on a screen 2.5 m away from the double slit.



(a) Estimate the fringe separation.

(b) Estimate the slit separation.

→ E-version in OUP Exercise Platform is also available.

☆ HKDSE 2016 Paper 1B Q6(a)

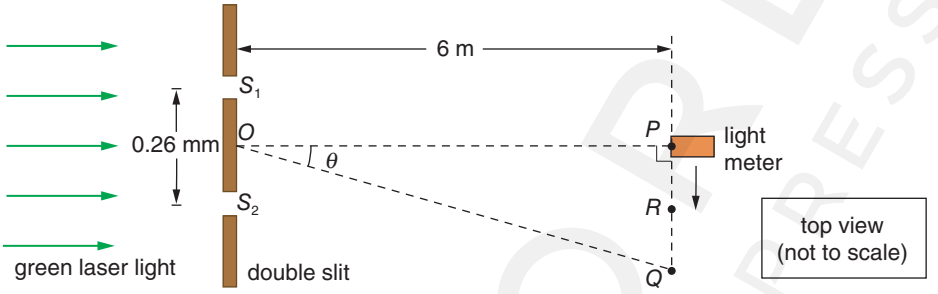
→ This video demonstrates how this question is solved.

Video

Ext

DSE practice 1 | Fringe separation and path difference

In the experiment below, a green laser beam of wavelength 520 nm passes through a double slit of slit separation 0.26 mm. A light meter, which measures the amount of light that it receives, is placed at P which is 6 m from the double slit and is equidistant from slits S_1 and S_2. The light meter is then moved from P towards Q. As the light meter moves, it records minima and maxima alternately.



(a) The light meter records the first maximum at P and the fourth maximum at Q. Find the separation between P and Q. (2 marks)

(b) Find the angle θ. (2 marks)

(c) The light meter records the second minimum at R. Find the path difference at R from S_1 and S_2. (2 marks)

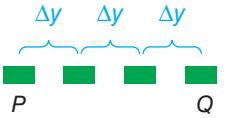
(d) Alternate minima and maxima can be detected along PQ only if the slits S_1 and S_2 are very narrow. Explain briefly. (2 marks)

Solution

(a) $\Delta y = \dfrac{\lambda D}{a} = \dfrac{(520 \times 10^{-9})(6)}{0.26 \times 10^{-3}} = 0.012$ m 1M

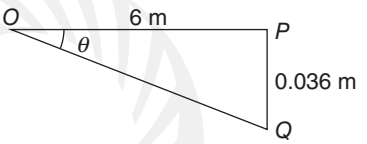
Separation between P and $Q = 3(\Delta y)$
$= 3(0.012)$
$= 0.036$ m 1A

Common mistake
Some students make mistakes in counting the number of Δy.



(b) $\tan\theta = \dfrac{0.036}{6}$ 1M

$\theta = 0.344°$ 1A



(c) Path difference at $R = 1.5\lambda$ 1M
$= 1.5(520)$
$= 780$ nm 1A

(d) Only narrow slits allow light waves to diffract enough 1A
so that the diffracted waves can overlap. 1A

Common mistake
Some students mix up slit width and slit separation. Some cannot state clearly that the diffracted waves must overlap for interference to occur.

▶ Revision 6 Q33(a) (p.148)

Exercise 6.2

★1 When monochromatic light passes through a double slit, bright fringes are observed. Which of the following combinations would produce bright fringes with the largest fringe separation?

	Slit separation / mm	Colour of light used
A	0.25	Red
B	0.25	Blue
C	0.5	Red
D	0.5	Blue

★2 When monochromatic light passes through a double slit, an interference pattern is produced on a screen. If the wavelength of the light is λ, what is the path difference at the nth dark fringe (counted from the central bright fringe) from the two slits?

A $(n - 0.5)\lambda$

B $n\lambda$

C $(n + 0.5)\lambda$

D $(n + 1)\lambda$

★3 A single light source and a double slit are used to produce an interference pattern (Fig a).

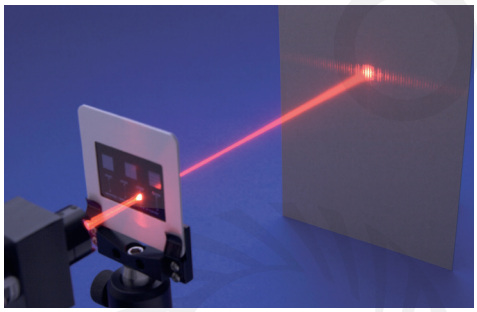


Fig a

(a) Explain whether we can replace the single light source and the double slit with two separate light sources to obtain a clear interference pattern.

(b) How will the fringe separation change if

(i) a green light beam is used instead of a red light beam;

(ii) a double slit with a larger slit separation is used.

★4 Ext In a double slit experiment using a monochromatic light source, bright fringes are formed on a screen 1.2 m away from the double slit (Fig b). The slit separation is 6.5×10^{-4} m.

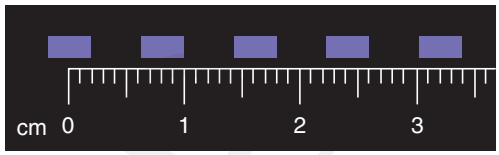


Fig b

(a) What is the wavelength of the light?

(b) If the slit-screen distance is changed so that the separation between adjacent fringes is doubled, what is the new slit-screen distance?

★5 Ext Monochromatic light passes through a double slit and forms an interference pattern on a screen. The wavelength of the light is 500 nm. The slit separation is 0.4 mm and the distance between the screen and the double slit is 1.6 m.

(a) Find the separation between the 4th-order bright fringe and the central bright fringe.

(b) What is the path difference from the two slits at the position where the 4th-order bright fringe is formed?

(c) In Figure c, sketch the interference pattern up to the 4th-order bright fringe.

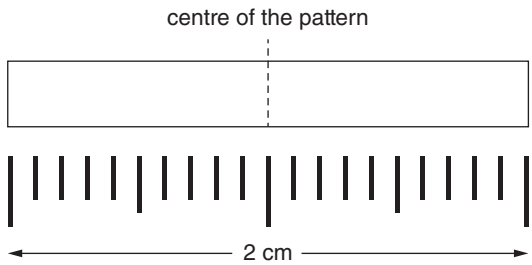


Fig c

★6 Figure d shows the interference pattern formed by directing red light onto a double slit.

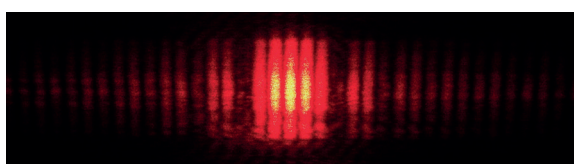


Fig d

(a) How are the bright fringes formed?

(b) Suggest a way of increasing the separation between the bright fringes.

6.3 Plane transmission grating

A **plane transmission grating** is a type of **diffraction grating**. It is a small piece of glass or plastic consisting of many evenly separated opaque parallel lines (Fig 6.3a). The space between two adjacent lines acts as a slit, so a grating has a lot of slits. A grating with more lines per mm is called a fine grating, while that with fewer lines per mm is called a coarse grating.

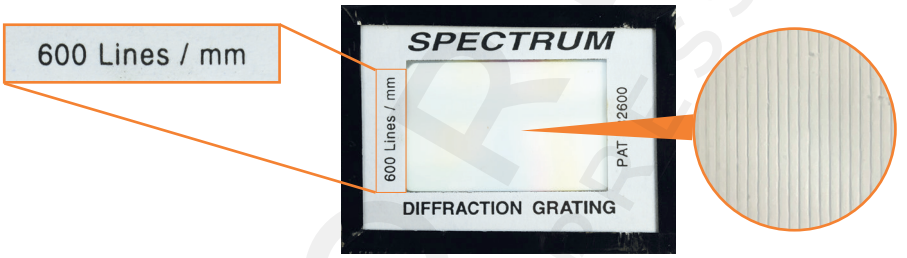


Fig 6.3a *Diffraction grating with 600 lines per mm, i.e. 600 slits in each millimetre.*

DSE goal

Examine the interference patterns in plane transmission grating.

DSE exam

23(1A)Q18

Video

→ This video demonstrates Expt 6c.

You must be very careful when using laser in experiments. Make sure you follow all the safety guidelines on p.102.

→ This is a Video Quiz about this experiment.

1 Examining the pattern

Experiment 6c / Patterns formed by gratings

1 Direct a laser beam towards a coarse grating. Capture the pattern on a screen.



2 Repeat step 1 with a fine grating.

3 Repeat step 1 with laser beams of different colours.

4 Alternatively, replace the laser with a ray box. View the filament through a coarse grating and a fine grating in turn.

Discussion

1 How does the pattern differ from that formed by a double slit?

2 How does the pattern change when
   (a) a fine grating is used;
   (b) the colour of the laser beam is changed?

plane transmission grating 平面透射光柵　　diffraction grating 衍射光柵

When a light wave passes through a diffraction grating, diffraction occurs at every slit. In certain directions, the diffracted waves from every slit undergo constructive interference and form bright fringes on the screen (Fig 6.3b).

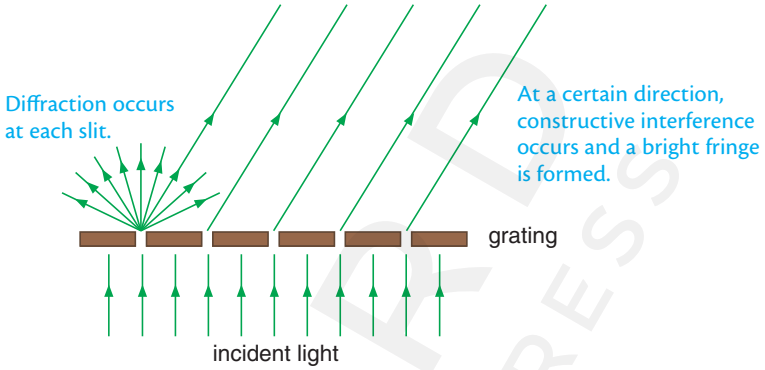


Fig 6.3b *Bright fringes formed in directions when diffracted waves from every slit undergo constructive interference.*

Figure 6.3c shows the interference pattern formed by a diffraction grating. It has the following features:

1 It is **symmetric** about the central bright fringe.

2 The **further away** from the central bright fringe, the **larger** the fringe separation.

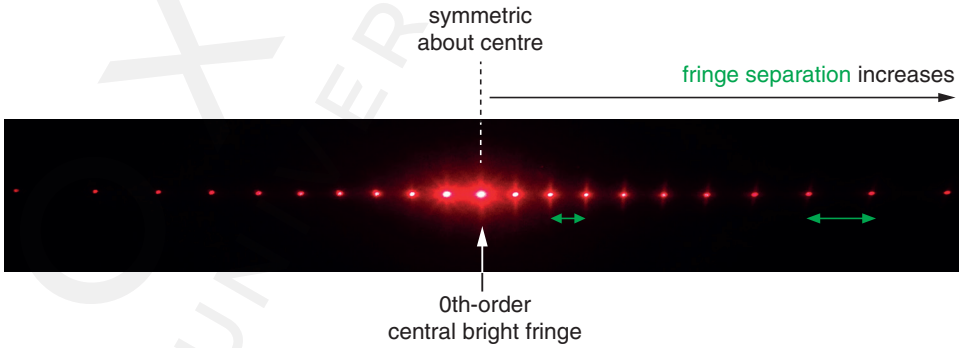


Fig 6.3c *Interference pattern formed by a grating.*

The pattern formed by a grating is about a hundred times wider than that formed by a double slit.

3 It is much wider than the interference pattern formed by a double slit (Fig 6.3d).

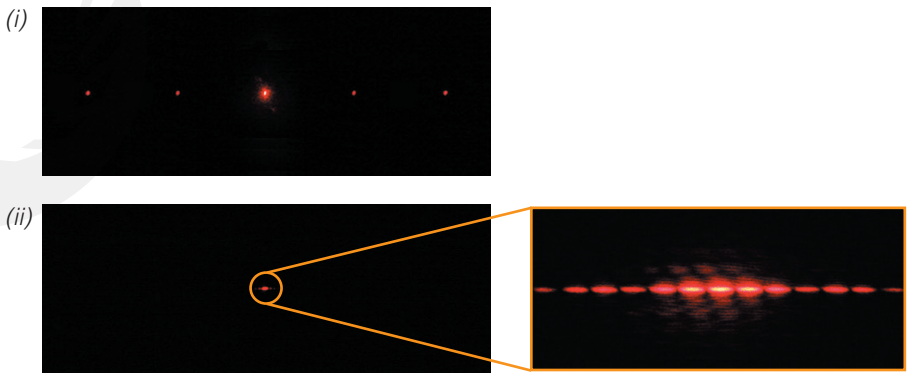


Fig 6.3d *Interference patterns formed by (i) a grating and (ii) a double slit at the same distance.*

2 Factors affecting the pattern

The interference pattern formed by a transmission grating is affected by the following factors.

a Grating spacing

The grating spacing d refers to the separation between two adjacent slits (Fig 6.3e). For a grating with 600 lines per mm,

$$d = \frac{1 \times 10^{-3}}{600} = 1.67 \times 10^{-6} \text{ m}$$

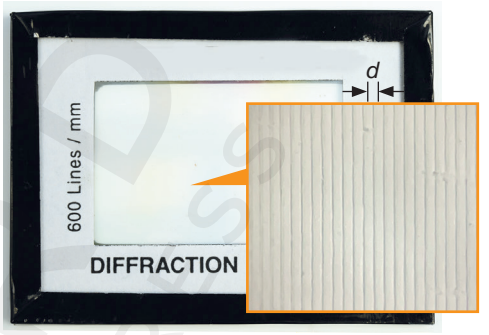


Fig 6.3e Grating spacing.

A grating with a smaller grating spacing produces fringes with a larger fringe separation (Fig 6.3f).

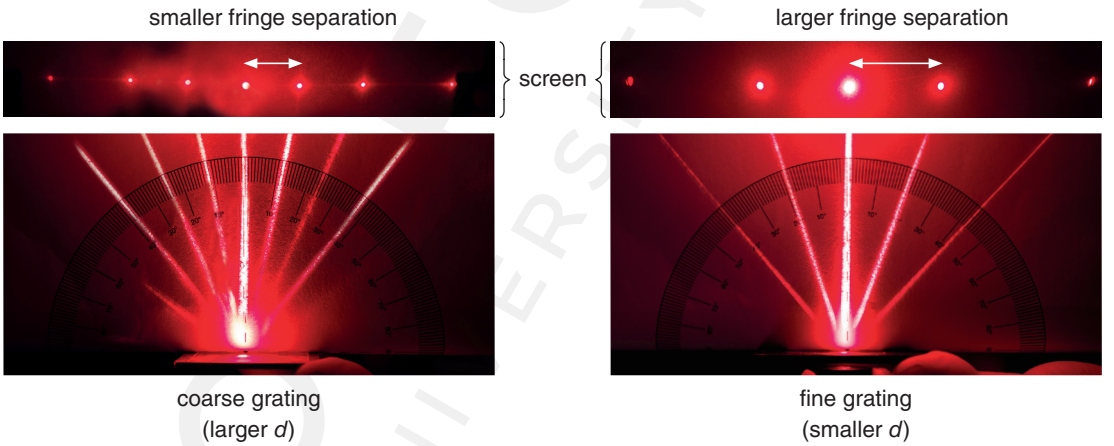


Fig 6.3f Bright fringes formed by gratings of different grating spacings.

b Wavelength

Light with a longer wavelength produces fringes with a larger fringe separation (Fig 6.3g).

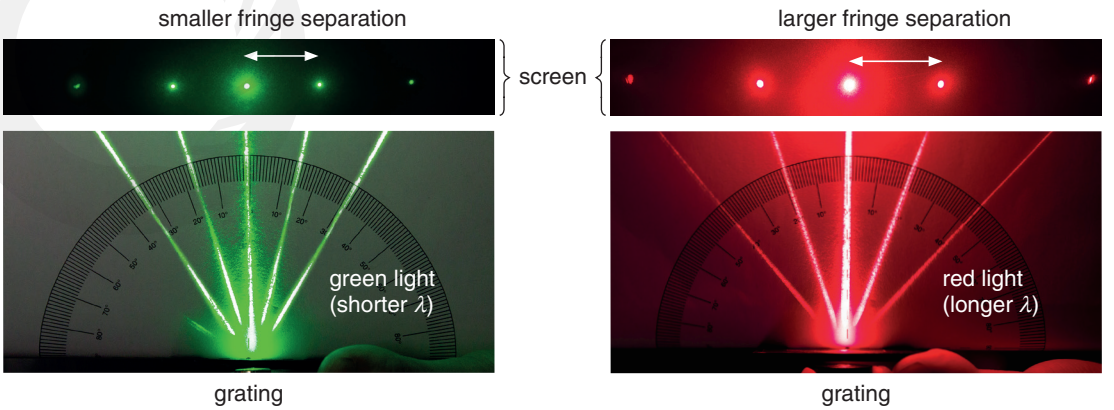


Fig 6.3g Bright fringes formed by different colours of light.

c Grating-screen distance

A larger distance D between the grating and the screen produces fringes with a larger fringe separation (Fig 6.3h).

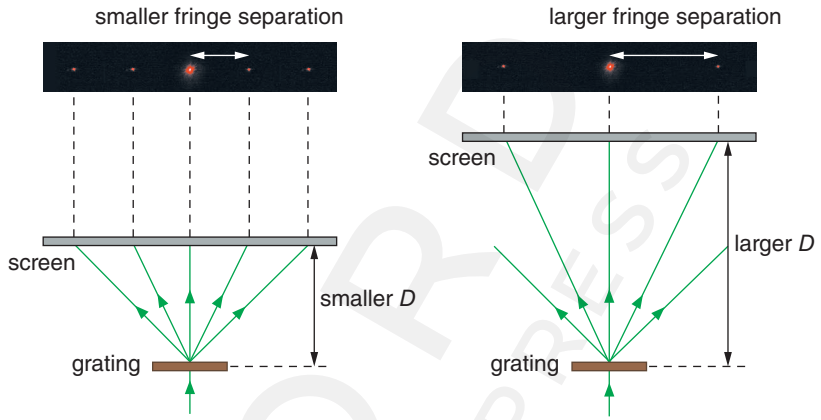


Fig 6.3h *Bright fringes formed with different grating-screen distances.*

In the interference pattern formed by a diffraction grating, the fringe separation increases when:

1 the grating spacing d decreases;

2 the wavelength λ of light increases;

3 the distance D between the grating and the screen increases.

Google Forms Microsoft Forms

Quick check 5

→ E-version in OUP Exercise Platform is also available.

1 A laser beam is directed towards a diffraction grating and an interference pattern is formed. Determine whether the following statements are true.

(a) The interference pattern is symmetrical about the central bright fringe. (T/F)

(b) The fringe separation remains unchanged. (T/F)

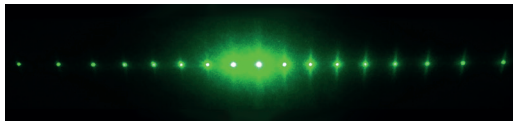
(c) The interference pattern formed is much wider than that formed by a double slit. (T/F)

2 A grating has 200 lines per mm. Find its grating spacing.

A 5×10^{-6} m B 5×10^{-3} m

C 200 mm D 2×10^{5} m

3 Fringes of green light are formed by a diffraction grating on a screen.



What can be done to obtain fringes which are more widely spaced? Put a tick in the corresponding box(es).

☐ Replace the grating with a double slit.

☐ Reduce the separation between the screen and the grating.

☐ Change the light source from green to red.

☐ Change to a grating with more lines per mm.

Ext

DSE goal

Apply $d \sin\theta = n\lambda$ to solve problems.

DSE exam

13(1A)Q23, 15(1A)Q17, 16(1B)Q6(b), 17(1A)Q20, 20(1A)Q14, 21(1A)Q17, 22(1A)Q18, 23(1A)Q18, etc.

3 The angle of diffraction of a fringe

Consider monochromatic light of wavelength λ being directed towards a diffraction grating of grating spacing d (Fig 6.3i). Bright fringes are formed on a screen which is at a distance D from the grating.

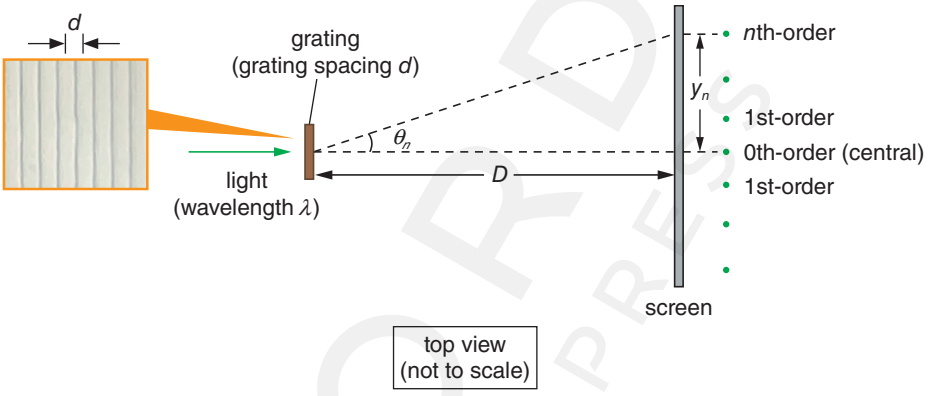
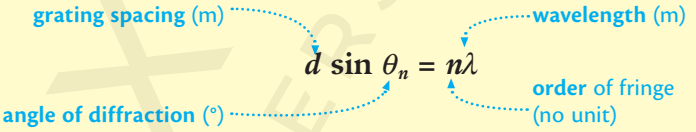


Fig 6.3i *Fringes formed when light passes through a grating.*

When ***D* is much larger than *d***, the **angle of diffraction** (or the *angular position*) θ_n of the nth-order bright fringe is given by



Note that:

1 The angle of diffraction θ_n of the nth-order fringe increases when

(a) the grating spacing d decreases;

(b) the wavelength λ of the light increases.

2 The distance y_n of the nth-order bright fringe from the central bright fringe is given by:

$$y_n = D \tan\theta_n$$

3 The fringe separation Δy is **not** constant. In addition, $\Delta y = \dfrac{\lambda D}{d}$ is **not** applicable since θ_n **is not small**.

4 By $d \sin\theta_n = n\lambda$ and $\sin\theta_n \leq 1$, the **highest order** of bright fringe is equal to the **largest integer** n satisfying

$$n \leq \frac{d}{\lambda}$$

Know more

Deriving $d \sin\theta_n = n\lambda$

We can derive this formula.

▶ p.150

Caution

θ_n is independent of D.

Know more

Different wording

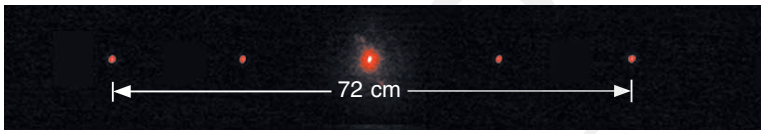
In some questions, 'nth-order diffracted light/beam' or 'nth-order maximum' may be used. Both correspond to 'nth-order bright fringe'.

angular position 角位置

Ext

Example 2 Fringes formed by a grating

Light of wavelength 630 nm is directed towards a diffraction grating. The figure shows the pattern projected on a screen 1 m from the grating. The distance between the two second-order bright fringes is 72 cm.



(a) Find the angle of diffraction θ_2 of the second-order bright fringe.

(b) Find the grating spacing d.

(c) Find the highest order of bright fringe.

(d) Find the total number of bright fringes formed.

Solution

(a) Distance of 2nd-order fringe from 0th-order fringe $= \frac{0.72}{2} = 0.36$ m

$\tan \theta_2 = \frac{0.36}{1}$

$\theta_2 = 19.8°$

(b) By $d \sin \theta_n = n\lambda$,

$d \sin 19.8° = (2)(630 \times 10^{-9})$

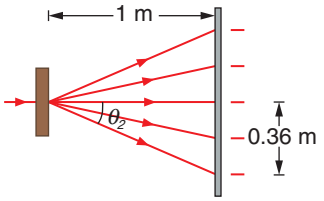
$d = 3.72 \times 10^{-6}$ m

(c) $n \leq \frac{d}{\lambda} = \frac{3.72 \times 10^{-6}}{630 \times 10^{-9}} = 5.9$

$\therefore$ The highest order of bright fringe is 5.

(d) Total number of bright fringes $= 5 \times 2 + 1 = 11$

Tip



The angle of diffraction of the 1st-order fringe is not 19.8°/2. Do you know how to find this angle?

Caution

Remember to count the 0th-order bright fringe.

Quick check 6

→ E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

1 Determine whether the following statements about the interference pattern formed by a grating are true.

(a) A bright fringe is always formed at $\theta_n = 0°$. (T/F)

(b) If red light and violet light pass through the same grating, the angle of diffraction of the 1st-order maximum of the red light is larger than that of the violet light. (T/F)

(c) The maximum number of fringes formed will increase if the grating-screen distance increases. (T/F)

2 Complete the following table.

	d / 10^{-6} m	θ_n	n	λ / 10^{-7} m
(a)	2		3	5
(b)	5	20°	4	

Ext

Learn by Practice 2

1 Laser light passes through a grating with 500 lines per mm. The angle of diffraction of the 3rd-order bright fringe is 37°. A screen is 0.5 m from the grating.

(a) Find the distance between the 3rd-order and 0th-order bright fringes on the screen.

(b) Find the wavelength of the light.

(c) Find the angle of diffraction of the 1st-order bright fringe.

2 Blue light of wavelength 460 nm passes through a coarse grating of grating spacing 10^{-5} m.

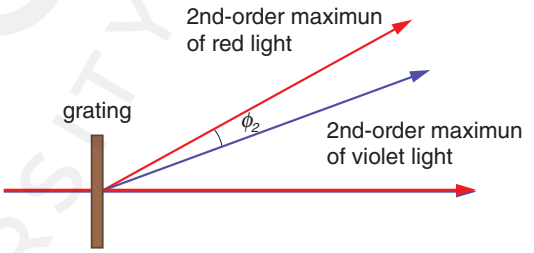
(a) Find the highest order of bright fringe.

(b) Find the angle of diffraction of the highest order bright fringe.

(c) Find the angle of diffraction of the highest order bright fringe if light of wavelength 600 nm is used.

Example 3 Multi-coloured interference pattern

Red light of wavelength 700 nm and violet light of wavelength 400 nm pass through a diffraction grating of 200 lines per mm.



(a) Find the angle of diffraction of the 2nd-order maxima (i.e. 2nd-order bright fringes) of the two lights.

(b) Find the angle of diffraction of the 3rd-order maxima of the two lights.

(c) Find the angle ϕ_2 between the 2nd-order maxima of the two lights.

(d) Find the angle ϕ_3 between the 3rd-order maxima of the two lights.

Solution

Grating spacing $d = \dfrac{1 \times 10^{-3}}{200} = 5 \times 10^{-6}$ m

Apply $d \sin \theta_n = n\lambda$.

2nd-order maxima	3rd-order maxima
(a) For red light, $(5 \times 10^{-6}) \sin \theta_{2R} = (2)(700 \times 10^{-9})$ $\theta_{2R} = 16.3°$ For violet light, $(5 \times 10^{-6}) \sin \theta_{2V} = (2)(400 \times 10^{-9})$ $\theta_{2V} = 9.21°$	(b) For red light, $(5 \times 10^{-6}) \sin \theta_{3R} = (3)(700 \times 10^{-9})$ $\theta_{3R} = 24.8°$ For violet light, $(5 \times 10^{-6}) \sin \theta_{3V} = (3)(400 \times 10^{-9})$ $\theta_{3V} = 13.9°$
(c) $\phi_2 = 16.3° - 9.21°$ $= 7.2°$	(d) $\phi_3 = 24.8° - 13.9°$ $= 10.9°$

▶ Exercise 6.3 Q5 (p.128)

Ext

White light is a mixture of lights of different wavelengths. What will be the interference pattern formed if a beam of white light is directed towards a diffraction grating?

Figure 6.3j shows the resulting pattern. The colour spectra indicate that different colours form maxima at different angles.

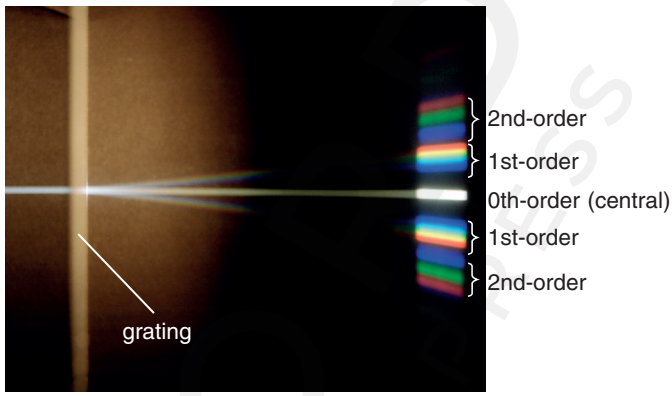


Fig 6.3j *Interference pattern formed from white light.*

The pattern has the following features:

All colours form maxima at the centre.

▶ **1** The central maximum is white.

In Example 3 on p.122, $\theta_{2R} > \theta_{2V}$ and $\theta_{3R} > \theta_{3V}$

▶ **2** For other orders of maxima, colour spectra are formed. For each spectrum, the **inner** edge is **violet** while the **outer** edge is **red**.

In Example 3, $\phi_3 > \phi_2$

▶ **3** The higher the order, the wider the spectrum.

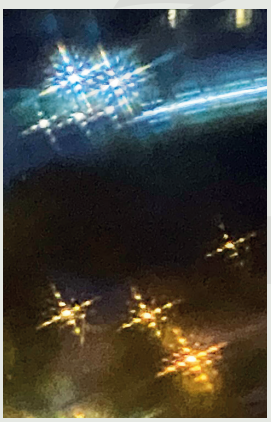
In Example 3, $\theta_{2R} > \theta_{3V}$

▶ **4** Adjacent spectra may overlap.

Know more

Grating in daily life

A gauze curtain can act like a diffraction grating. You can see interference patterns through it.

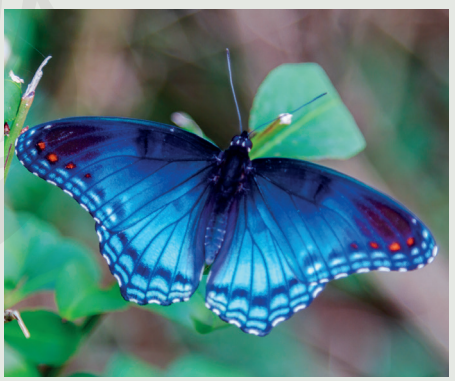


▶ p.152

Colours produced by interference

Butterfly wings are colourful. Their colours are not produced by pigments but by the interference of light.

The colours reflected by a DVD are produced by a similar effect.





▶ p.151

Ext

DSE exam
14(1B)Q7

4 Estimating the wavelength

When you look at a small monochromatic light source (e.g. an LED) through a grating, you will see the maxima at different directions (Fig 6.3k). This allows us to measure the wavelength of the light.

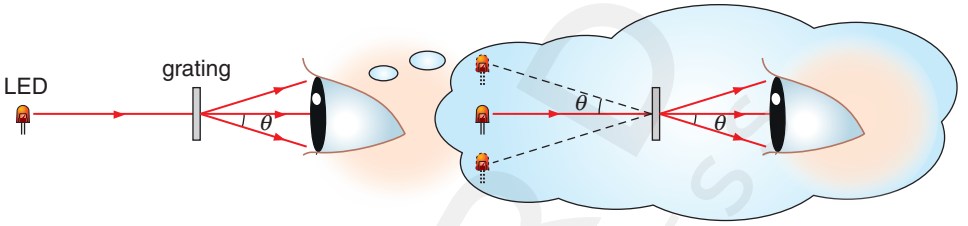


Fig 6.3k *Look at an LED through a grating.*

Video

→ This video shows how to estimate the wavelength of light using a grating.

Example 4 Estimating wavelength using a grating

Fanny looks at a red LED through a diffraction grating with grating spacing 4.35×10^{-6} m and sees several bright fringes (Fig a). She asks her classmate to put a pen at the position of a bright fringe and measures the distance y between the fringe and the LED. The angle θ_n is then calculated. They repeat the measurement for several bright fringes and plot a graph of $\sin \theta_n$ against the order of fringe n (Fig b). What is the wavelength λ of the light?

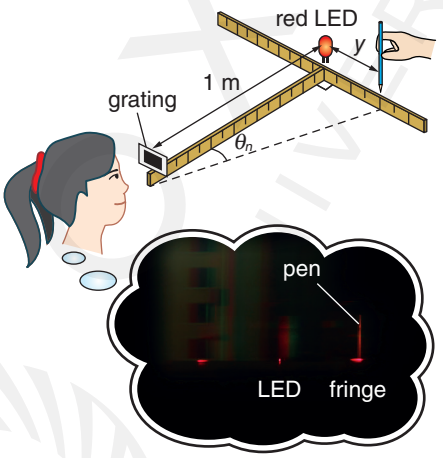


Fig a

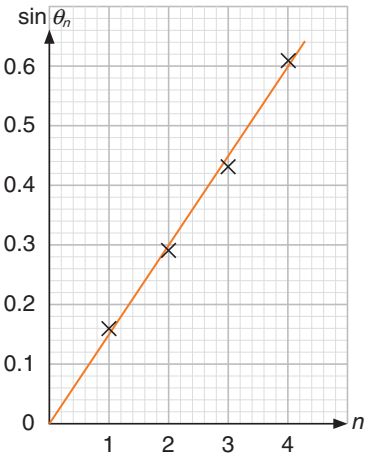


Fig b

Solution

$\tan \theta = \frac{y}{1}$ ▶

By $d \sin \theta_n = n\lambda$, $\sin \theta_n = \frac{\lambda}{d} \times n$

Compare with the equation of a straight line:
$y = mx + c$

$\therefore$ Slope of graph $= \frac{\lambda}{d}$

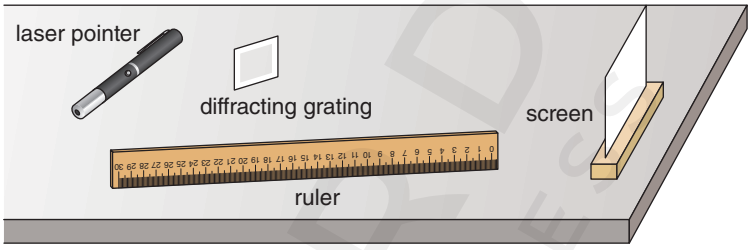
$\therefore \lambda = d \times$ slope of graph $= 4.35 \times 10^{-6} \times \frac{0.6 - 0}{4 - 0} = 6.53 \times 10^{-7}$ m

▶ Exercise 6.3 Q9 (p.129)

Ext

Writing practice

You are given a laser pointer, a diffraction grating with grating spacing d, a screen and a ruler.



Question 1

Describe how you would use the above apparatus to estimate the wavelength of the laser light. (4 marks)

Answer

Direct the laser beam perpendicularly to the grating to project an interference pattern on the screen. 1A

Measure the distance D between the grating and the screen. 1A

Measure the distance y between the zeroth-order and the nth-order bright fringes. 1A

Calculate the angle of diffraction θ_n by $\tan \theta_n = \frac{y}{D}$ and then calculate the wavelength of light λ by $d \sin \theta_n = n\lambda$. 1A

Question 2

Suggest and explain one precaution that can make the results more accurate. (2 marks)

Answer

Using higher-order bright fringe rather than the first-order bright fringe when measuring y. 1A

This reduces the percentage error in y. 1A

Or ▶

Measure the values of y using fringes of the same order on both sides and take the average of the two values obtained. 1A

This reduces the error due to the alignment of the screen, the grating and the laser beam. 1A

Practise yourself

You are given a laser pointer, a double slit with slit separation a, a screen and a ruler.

1 Describe how you would use the above apparatus to estimate the wavelength of the laser light. (4 marks)

2 Suggest and explain one precaution that can make the results more accurate. (2 marks)

Extd

DSE exam
19(1A)Q19

5 Comparing double slit and grating

Both a double slit and a diffraction grating can form interference patterns. The patterns have different features.

1 A grating allows more light to pass through than a double slit and so the fringes formed by a grating are brighter.

2 Refer to the two formulae:

$a \sin \theta_n = n\lambda$ (double slit)

$d \sin \theta_n = n\lambda$ (grating)

The slit separation a is about 10^{-3} m, while the grating spacing d is about 10^{-6} m. Since $d << a$, for the same order of fringe, the value of θ_n is much larger for a grating. That is, the fringes formed by a grating are more widely spaced.

3 The highest order of fringe formed by a grating is given by the largest integer n satisfying

$$n \leq \frac{d}{\lambda}$$

However, the highest order of fringe formed by a double slit cannot be found using a similar method. The reason is out of the scope of this book.

Table 6.3a gives a detailed comparison between the two patterns.

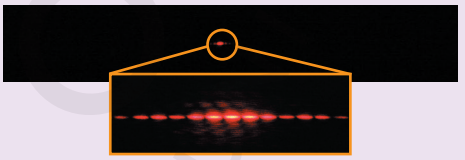
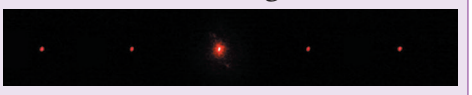
	Double slit	Grating
		
Brightness of pattern	dimmer	brighter
Span of pattern	narrower	wider
Angle of diffraction θ_n of the nth-order bright fringe	for $D >> a$, $a \sin \theta_n = n\lambda$	for $D >> d$, $d \sin \theta_n = n\lambda$
Fringe separation Δy	for $D >> a$ and very small θ_n, Δy unchanged for the entire pattern (fringes evenly spaced) $\Delta y = \frac{\lambda D}{a}$	is larger for fringes further from the centre (larger n)
Distance between nth-order and 0th-order bright fringes	$y_n = n \times \Delta y$	$y_n = D \tan \theta_n$
Highest order of bright fringe	–	largest integer n satisfying $n \leq \frac{d}{\lambda}$

Table 6.3a *Comparison between the interference patterns produced by a double slit and a grating.*

Ext

☆ HKDSE 2016 Paper 1B Q6(b)

→ This video demonstrates how this question is solved.

Video

DSE practice 2 | Double slit and diffraction grating

(a) Laser light of wavelength 650 nm passes through double slit X with silt separation 0.5 mm. An interference pattern is formed on a screen 1.5 m away from the double slit.

(i) Find the separation between the 1st-order and 0th-order bright fringes on the screen. (2 marks)

(ii) Sketch the interference pattern up to the 2nd-order fringes. (2 marks)

(b) Double slit X is replaced by diffraction grating Y with 200 lines per mm.

(i) Find the separation between the 1st-order and 0th-order bright fringes on the screen. (3 marks)

(ii) Sketch the interference pattern up to the 2nd-order fringes. (2 marks)

(iii) A laser emitting light of wavelength 500 nm and diffraction grating Z with 500 lines per mm are also available. Among the two lasers and two gratings, which combination will produce a pattern with the largest fringe separation? (1 mark)

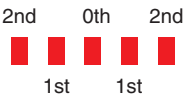
Solution

(a) (i) $\Delta y = \dfrac{\lambda D}{a}$

$= \dfrac{(650 \times 10^{-9})(1.5)}{0.5 \times 10^{-3}}$ 1M

$= 1.95 \times 10^{-3}$ m (= 1.95 mm) 1A

(ii)



(Evenly spaced) 1A

(Symmetry about 0th-order) 1A

(b) (i) By $d \sin \theta_n = n\lambda$,

$\dfrac{1 \times 10^{-3}}{200} \sin \theta_1 = (1)\ (650 \times 10^{-9})$ 1M

$\theta_1 = 7.47°$

The separation $= D \tan \theta_1$

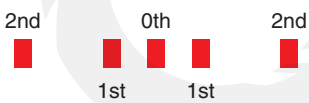
$= (1.5) \tan 7.47°$ 1M

$= 0.197$ m 1A

Common mistake
Some students wrongly use $\Delta y = \dfrac{\lambda D}{d}$.

This is about 100 times the answer to (a)(i).

(ii)



(Separation increases when n increases) 1A

(Symmetry about 0th-order) 1A

(iii) Light of wavelength 650 nm and grating Z. 1A

Common mistake
Some students wrongly draw evenly spaced fringes. Some miss out the 0th-order. Some wrongly draw patterns that are not symmetrical about the 0th-order.

▶ Revision 6 Q33(b) (p.148)

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Exercise 6.3

1 When light passes a diffraction grating, bright fringes are formed on the screen behind the grating. Which of the following wave phenomena does light display in this case?

(1) Refraction (2) Diffraction

(3) Interference

A (2) only B (1) and (3) only

C (2) and (3) only D (1), (2) and (3)

***2** When monochromatic light passes through a diffraction grating, bright fringes are observed on a screen. Which of the following combinations would produce bright fringes with the largest separation on the screen?

	No. of lines per cm in grating	Colour of light
A	3000	Red
B	3000	Blue
C	5000	Red
D	5000	Blue

***3** Ext Monochromatic light is directed onto a diffraction grating. The 2nd-order bright fringe is formed at an angular position θ. Its distance from the central bright fringe is d on the screen (Fig a).

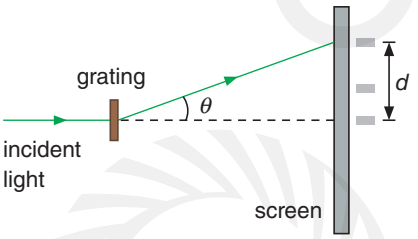


Fig a

If a diffraction grating of half the grating spacing is used, which of the following would happen?

A The 2nd-order bright fringe would be formed at the angular position $\frac{1}{2}\theta$.

B The 2nd-order bright fringe would be formed at the angular position 2θ.

C The 2nd-order bright fringe would be $2d$ from the central bright fringe on the screen.

D The 1st-order bright fringe would be formed at the angular position θ.

***4** A diffraction grating is used to produce an interference pattern of light on a screen behind it. Which of the following changes would produce a pattern with larger fringe separation?

(1) Replacing the grating with the one of larger grating spacing.

(2) Replacing the grating with the one of more lines per mm.

(3) Replacing the grating with a double slit.

A (1) only B (2) only

C (1) and (3) only D (2) and (3) only

***5** Ext A source emits red and violet lights of wavelengths 700 nm and 400 nm respectively. These lights pass through a diffraction grating of 600 lines per mm. Bright fringes are formed on a screen 50 cm behind the grating.

(a) Find the angle of diffraction of the 1st-order maxima of the two coloured lights.

(b) Find the angle between the 1st-order maxima of the two coloured lights.

(c) Find the distance between the 1st-order maxima of the two coloured lights on the screen.

(d) Find the angle of diffraction of the highest order fringes for the two coloured lights.

***6** Ext White light from an LED lamp is incident on a device X as shown (Fig b). Coloured fringes are formed on the wall 30 cm behind X.

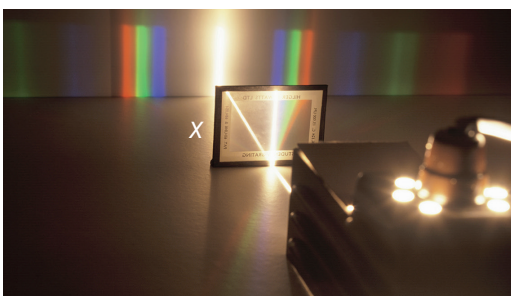


Fig b

(a) Is X a double slit or a diffraction grating? Explain your answer.

(b) Suggest a way of increasing the width of the 2nd-order spectrum.

***7** A laser beam passes through a diffraction grating and an interference pattern is formed on a screen. How will the interference pattern change if the whole set-up is put underwater? Explain.

★8 Ext A beam of monochromatic light of wavelength 520 nm is directed onto a diffraction grating with 250 lines per mm. A diffracted beam is found at an angle of 23° to the incident direction.

(a) What is the order of the diffracted beam?

(b) Show that this is not the highest order of diffracted beam.

★9 Ext Mary uses the setup as shown (Fig c) to estimate the range of wavelength of the light emitted by a small light bulb. She looks at the bulb through a diffraction grating and sees many images of the bulb. A vertical pen P is first moved along meter rule B until it is in line with end X of the 1st-order diffracted image of the bulb (Fig d). The distance L between the pen and the bulb is measured to find the diffraction angle θ.

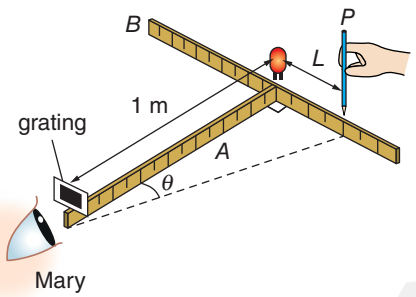


Fig c

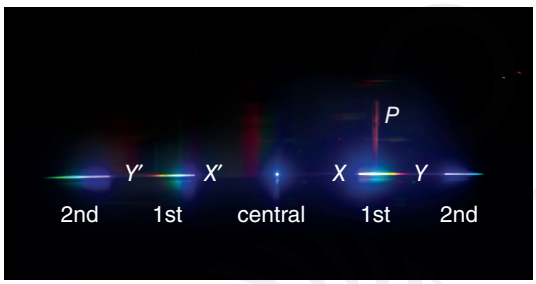


Fig d

(a) Which end, X or Y, corresponds to light of a shorter wavelength?

(b) The teacher asks Mary to measure the distance L' between X' and the central image, and then take the average value of L and L'. State the advantage of doing so.

(c) The teacher then asks Mary to repeat the measurement using the 2nd-order image. Which measurement, the 1st-order one or the 2nd-order one, will give a more accurate estimation of the wavelength of the light? Explain briefly.

★10 Ext Monochromatic light of wavelength 620 nm is incident on a diffraction grating (3000 lines per cm). Bright fringes are produced on a screen behind the grating.

(a) Find the highest order of fringe that can be observed.

(b) Suggest a way of increasing the maximum number of bright fringes produced.

(c) The screen can just capture the 2nd-order bright fringe (Fig e).

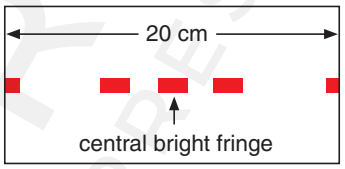


Fig e

(i) Estimate the grating-screen distance.

(ii) Estimate the distance between the 1st-order and 2nd-order bright fringes.

★11 Ext A student directs red light and green light towards a diffraction grating in turn. The grating has 200 lines per mm. Bright fringes are formed on a screen 20 cm from the grating (Fig f).

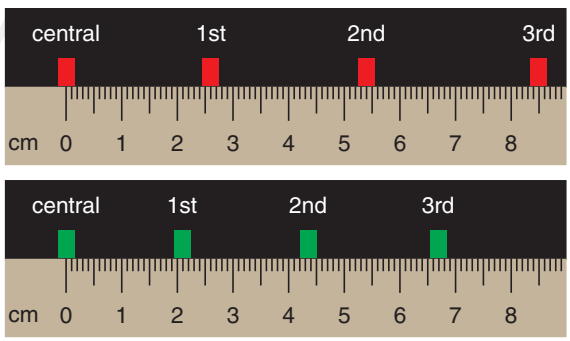


Fig f

(a) Estimate the wavelengths of the red and green lights.

(b) Estimate the highest order of bright fringe produced for each light.

(c) In the same graph, sketch how $\sin\theta$ varies with n for each light, where θ is the angle of diffraction of a bright fringe and n is the order of bright fringe.

6.4 Electromagnetic waves

DSE goal

State the relative positions of visible light and other parts of the EM spectrum.

DSE exam

15(1A)Q20, 21(1A)Q19, 23(1A)Q22

Flipped classroom

Watch a video and answer the questions.

1 Different kinds of electromagnetic waves

You must have heard of these terms: **radio waves**, **microwaves**, **infra-red radiation**, **ultra-violet radiation**, **X-rays** and **gamma rays**. They, like visible light, are electromagnetic waves (EM waves). They have different wavelengths. Figure 6.4a shows the whole **electromagnetic spectrum**.

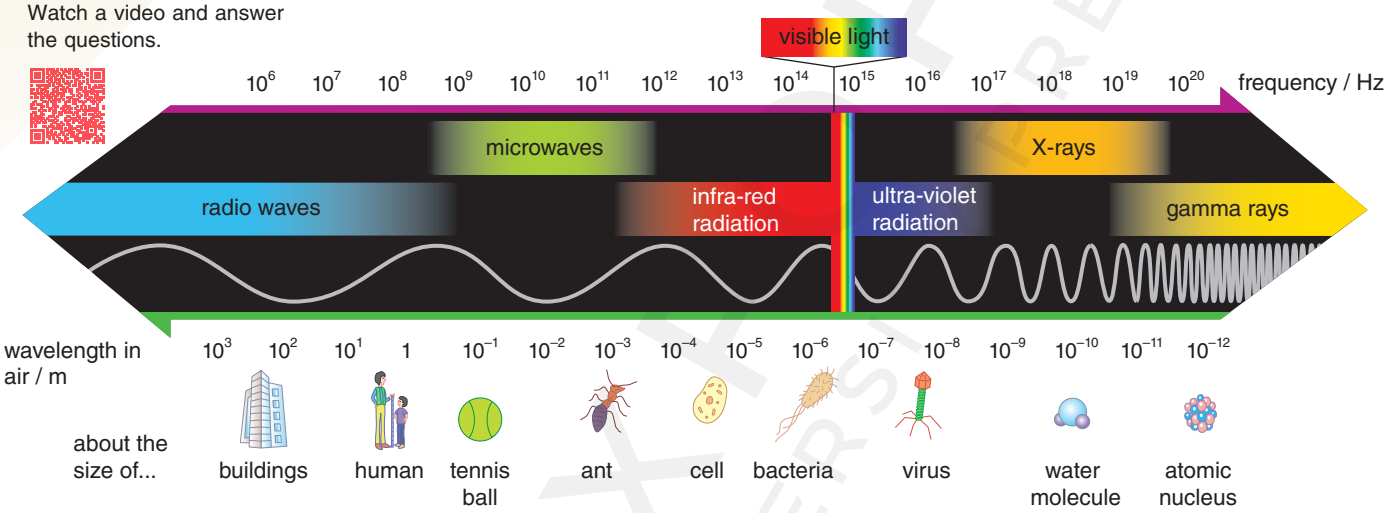


Fig 6.4a *The electromagnetic spectrum.*

Visible light only occupies a very small range on the spectrum. Other EM waves are invisible to us. Note that there are no clear-cut boundaries between adjacent kinds of EM waves.

a Radio waves

Radio waves have the longest wavelength (lowest frequency) in the EM spectrum. All EM waves with wavelengths longer than 10^{-1} m can be regarded as radio waves. They can diffract significantly around buildings, so they are widely used in signal transmission (Fig 6.4b). They are also used in contactless payment.



mobile phone signals (4G, 5G, Wi-Fi)



TV and radio broadcast



contactless payment

Fig 6.4b *Applications of radio waves.*

STSE

Radiation scare

Many mobile phone antennas are installed on residence rooftops. Would the radiation emitted by them affect our health? Search for information about this issue.



radio wave 無線電波　microwave 微波　infra-red radiation 紅外輻射　ultra-violet radiation 紫外輻射
X-ray X 射線　gamma ray 伽瑪射線　electromagnetic spectrum 電磁波譜

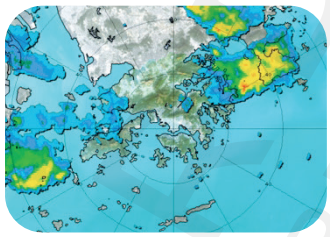
b Microwaves

Microwaves are sometimes regarded as short-wavelength radio waves. Their wavelengths range from about 10^{-3} m to 10^{-1} m. They are commonly used to heat food (Fig 6.4c). They are also used in radars to locate aircraft and rainstorms. As they can pass through the Earth's atmosphere, they are also used in satellite telecommunications.

Cross-link
We will learn more about microwave ovens in Book E3 Ch 1.2.



microwave oven



radar image



antenna for receiving satellite signals

Fig 6.4c *Applications of microwaves.*

Remind Ss that hot objects do not only emit IR. An object that is hot enough can emit visible light and other EM waves. The composition of EM waves emitted changes with temperature.

c Infra-red (IR) radiation

Infra-red radiation is next to visible light on the EM spectrum. Its wavelength ranges from about 7×10^{-7} m (the red edge of the visible spectrum) to 10^{-3} m. All objects emit IR radiation, but a hotter object emits more. It is widely used in heating, measuring temperature and signal transfer (Fig 6.4d).

Video

→ The video 'Thermographs' shows the images taken by an infra-red imaging camera.

→ The video 'An experiment showing the data transmission by infra-red radiation' shows the images taken by an infra-red imaging camera.

Heating

A halogen oven cooks food by emitting a large amount of IR radiation.





A radiant heater warms objects up by exposing them to IR radiation without heating up the surrounding air.

Measuring temperature

IR images can be used to monitor people's body temperature.





An IR thermometer works by detecting the IR emitted by a person.

Signal transfer

An automatic faucet has an IR sensor to detect hand motion for starting/stopping water flow.





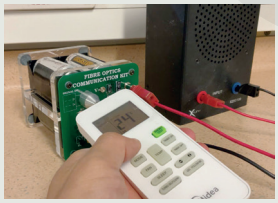
A remote control sends IR signals to operate a device.

Fig 6.4d *Applications of IR radiation.*

Know more

Seeing and hearing IR

We can see and hear IR by using suitable equipment.



▸ p.151

d Ultra-violet (UV) radiation

Ultra-violet radiation is on the other side of visible light on the EM spectrum. Its wavelength ranges from about 10^{-8} m to 4×10^{-7} m (the violet edge of the visible spectrum). It carries high energy. Excessive exposure to it can cause health risks such as cataract and skin cancer. Figure 6.4e shows some of its applications.

Mild exposure to UV radiation is essential to health. It stimulates the production of vitamin D in our body. ▶

✎ Washing powder also contains fluorescent chemicals that glow under UV radiation. This makes clothes look brighter in sunlight.



Medical instruments are sterilized in a UV sterilizer.



UV lamps are used for treating *jaundice* in new born babies.

Fig 6.4e *Applications of UV radiation.*

STSE

UV threat: Depletion of the ozone layer

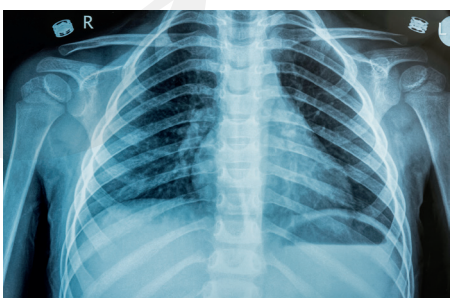
The atmosphere has protected us from most of the harmful UV radiation emitted from the Sun. However, recent human activities have threatened this function of the atmosphere.

▶ p.150

e X-rays

The wavelength of X-rays ranges from about 10^{-11} m to 10^{-8} m. They have high energy and high penetrating power. They can penetrate flesh but may be blocked by bones and metals (Fig 6.4f). A long-term or strong exposure to them can damage living cells deep in the body.

(i)



(ii)



Fig 6.4f *The penetrating power of X-rays makes them suitable to produce see-through images of (i) a human body and (ii) a luggage.*

jaundice 黃疸病

Cross-link

We will learn more about X-rays and gamma rays in Book 5.

Imaging using X-rays and gamma rays will be discussed in Book E4.

f Gamma rays

Gamma rays have the shortest wavelengths (shorter than 10^{-11} m) on the EM spectrum. They have the highest energy and highest penetrating power. They can damage cells deep in the body. Nevertheless, there are still important applications that make use of gamma rays (Fig 6.4g).

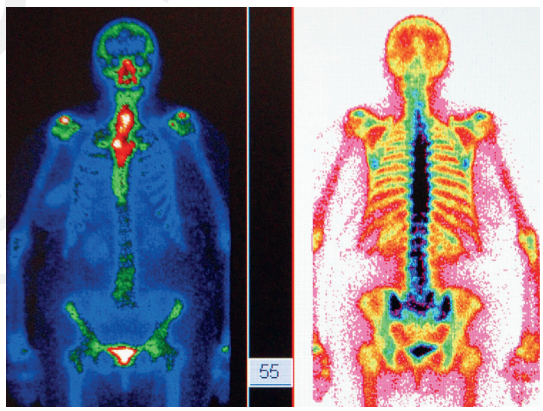
(i)



(ii)



(iii)



This logo is used to show that a food product has been treated with gamma rays.

Fig 6.4g *Gamma rays are used (i) in radiotherapy to kill cancer cells, (ii) to kill bacteria in food, and (iii) in medical imaging.*

Google Forms

Microsoft Forms

Quick check 7

→ E-version in OUP Exercise Platform is also available.

1 Listed below are different types of EM waves.

(1) Gamma rays, (2) infra-red radiation, (3) microwaves, (4) radio waves, (5) ultra-violet radiation, (6) visible light, (7) X-rays

Arrange them in ascending order of frequency.

A (1), (7), (5), (6), (2), (3), (4)

B (1), (5), (7), (6), (3), (2), (4)

C (4), (3), (2), (6), (5), (7), (1)

D (4), (2), (3), (6), (5), (1), (7)

2 Which of the following is/are application(s) of infra-red radiation?

☐ Radar

☐ Sterilization

☐ Measuring temperature

DSE goal

State that the speed of EM waves in a vacuum is 3.0×10^8 m s^{-1}.

DSE exam

13(1B)Q7, 19(1A)Q22

Cross-link

Electric and magnetic fields will be studied in Book 4.

2 Properties of electromagnetic waves

All EM waves have the following properties:

1 They are **transverse waves** consisting of varying electric and magnetic fields (Fig 6.4h).

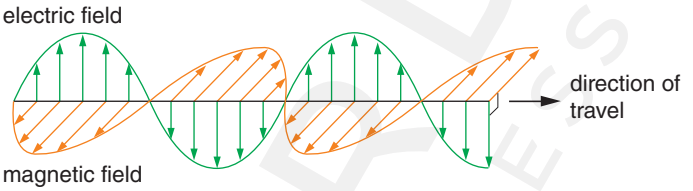


Fig 6.4h *Propagation of an EM wave.*

This video demonstrates the propagation of an EM wave.

Video

2 Unlike water waves, they do not require a medium to travel. In a **vacuum**, they all travel at a **constant speed c of 3.00×10^8 m s^{-1}**. They slow down when entering another medium from a vacuum.

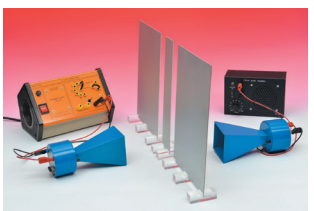
No waves or objects can travel faster than this speed, whether they are in a vacuum or any medium.

3 Like other waves, EM waves carry energy. Travelling EM waves transmit energy from one place to another.

4 They obey the relation $v = f\lambda$.

5 They exhibit reflection, refraction, diffraction and interference.

We can demonstrate these phenomena using microwaves.



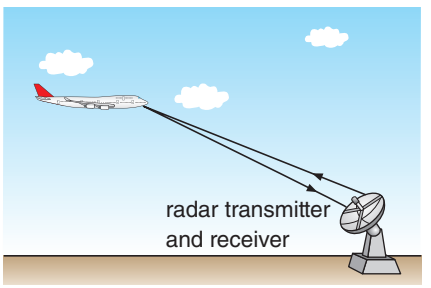
Video

→ This video shows the wave phenomena exhibited by microwaves.

Example 5 Radar

A radar system consists of a transmitter and a receiver. The transmitter sends out short pulses of microwaves. If the pulses hit an object such as an aircraft, some of the pulses bounce back to the receiver.

A radar pulse bounces back from a plane 4×10^{-5} s after it is transmitted. What is the distance of the plane from the radar station?



Solution

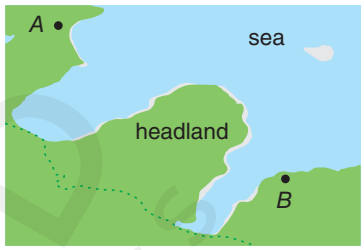
Since the speed of microwaves (3×10^8 m s^{-1}) is much higher than the speed of the plane (~250 m s^{-1}), the distance travelled by the plane during this time interval is negligible.

Time for pulse to travel from transmitter to plane $= \dfrac{4 \times 10^{-5}}{2} = 2 \times 10^{-5}$ s

Distance $= vt = (3 \times 10^8)(2 \times 10^{-5}) = 6000$ m

Example 6 Using walkie-talkies

In Hong Kong, there are two types of walkie-talkies for public use, one using 27-MHz radio waves and the other 409-MHz radio waves.



(a) Find the wavelengths of the two radio waves in air.

(b) Peter and Tony go hiking with walkie-talkies. As Peter reaches B as shown, he receives signals from Tony at A which is out of his sight.

(i) State the wave phenomenon which allows Peter to receive radio signals from Tony.

(ii) Which type of walkie–talkie has a better performance in this case? Explain briefly.

(iii) Explain why the reception is affected when a large cargo ship passes near Peter.

Solution

(a) Apply $v = f\lambda$.

Wavelength of 27-MHz radio waves $= \frac{v}{f} = \frac{3 \times 10^8}{27 \times 10^6} = 11.1$ m

Wavelength of 409-MHz radio waves $= \frac{v}{f} = \frac{3 \times 10^8}{409 \times 10^6} = 0.733$ m

(b) (i) Diffraction

(ii) The one using 27 MHz radio waves has a better performance since the radio waves have a longer wavelength and are diffracted more around the headland.

(iii) The radio waves reflected by the cargo ship interfere with the waves coming directly from Tony, so the reception is affected.

Google Forms Microsoft Forms

Quick check 8

1 Which of the following electromagnetic waves can form a clear interference pattern using a double slit with slit width 8 m?

A X-rays B Visible light

C Microwaves D Radio waves

2 All EM waves

- ☐ are transverse waves;
- ☐ travel at the same speed in a vacuum;
- ☐ travel at the same speed in any medium;
- ☐ can exhibit diffraction.
- ☐ carry energy.

Learn by Practice 3

1 State the range of wavelength of visible light, and the colour that has the longest wavelength.

2 A pulse of microwave is emitted from a radar station and its echo is received after 8.5×10^{-5} s. What is the distance between the object detected and the radar station?

3 The broadcasting frequency of RTHK Radio 1 ranges from 92.6 MHz to 94.4 MHz. Find the range of the broadcasting wavelength.

4 Find the frequencies of the following EM waves.

(a) Radio waves of wavelength 1 km

(b) Gamma rays of wavelength 3×10^{-12} m

→ E-version in OUP Exercise Platform is also available.

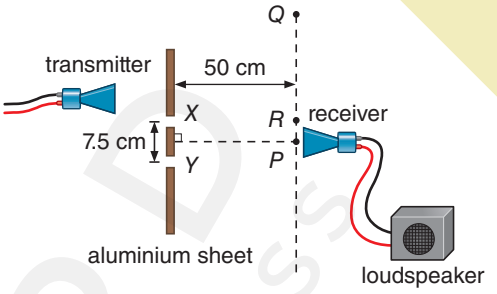
☆ HKDSE 2013 Paper 1B Q7

→ This video demonstrates how this question is solved.

Video

DSE practice 3 | Interference of microwaves

Steve wants to study the interference of microwaves. He places a 12-GHz microwave transmitter in front of an aluminium sheet with two slits X and Y separated by 7.5 cm. He then connects a receiver to a loudspeaker and moves it along PQ, which is 50 cm from XY. P is equidistant from X and Y.



(a) As the receivers moves from P to Q, the loudspeaker produces loud and soft sounds alternately. Explain why. (2 marks)

(b) What is the restriction on the widths of the slits in this experiment? (1 mark)

(c) Find the wavelength of the microwaves. (2 marks)

(d) Steve expects the loudspeaker to produce a loud sound when the receiver is at P. However, it does not. Explain briefly why this problem exists and suggest a method to fix it. (2 marks)

(e) After the problem in (d) is fixed, the receiver is moved from P to Q again. At most how many loud sounds can be produced? (2 marks)

Solution

(a) The path difference of the diffracted waves from X and Y to the receiver varies along PQ. 1A

Minima are recorded where destructive interference occurs, so soft sounds are produced. Maxima are recorded where constructive interference occurs, so loud sounds are produced. 1A

(b) They should be comparable to the wavelength of the microwaves. 1A

(c) By $v = f\lambda$,

$3 \times 10^8 = (12 \times 10^9)\lambda$ 1M

$\lambda = 0.025$ m (= 2.5 cm) 1A

Common mistake
Students may overlook that microwaves travel at the speed of light.

(d) The distances of the transmitter from X and Y are different, so the microwaves coming out from the two slits may not be in phase. 1A

The problem can be fixed by locating the transmitter at equal distance from X and Y. 1A

Common mistake
Students may not know that $\Delta <$ slit separation. Some may not be aware that the slit separation a is not negligible when compared with the distance D of the receiver from the double slit and so the equation $a \sin\theta = n\lambda$ cannot be applied. Some may forget to count the the 0-th order maximum.

(e) Path difference $\Delta < XY = 7.5$ cm $= 3\lambda$ 1M

$\therefore$ Loud sounds can be produced when $\Delta = 0\lambda$, 1λ and 2λ.

$\therefore$ At most 3 loud sounds can be produced. 1A

▶ Revision 6 Q30 (p.146)

Nature of science

Discovery of DNA structure by X-ray diffraction

In the early 1950s, many scientists were competing to be the first to figure out the structure of the DNA molecule—the blueprint of life. Different models were proposed by different research teams but none were widely accepted then.

In 1952, *Rosalind Franklin* took an X-ray diffraction photo of DNA. *James Watson* and *Francis Crick* at the Cavendish laboratory, Cambridge, got hold of the photo through *Maurice Wilkins*, a colleague of Franklin at King's College. After studying the diffraction pattern on the photo, in 1953 Watson and Crick managed to build a correct model of the DNA molecule with a double helical structure.

For their remarkable work in proposing the double helical structure of DNA, Watson and Crick shared the 1962 Nobel Prize for Physiology and Medicine with Wilkins.

Although Franklin had made a crucial contribution to the cracking of the secret of life, she did not share this Nobel Prize as she died of cancer in 1958. The rules do not allow a Nobel Prize to be awarded to a person who has passed away.

We can learn about the nature of science through the story of the discovery of the structure of DNA:

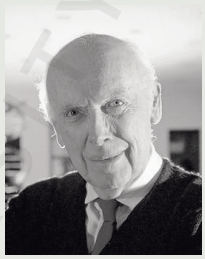
- Scientists often attempt to explain what they observe through building scientific models.
- Some scientists focus on experimental work while others focus on model-building. Often scientific advances are a result of sharing findings and collaboration among scientists (though in this case, Franklin had not formally agreed to share her data).
- Science (interference of waves) and technology (X-ray crystallography) are interrelated.



Rosalind Franklin (1920–1958).



Maurice Wilkins (1916–2004).



James Watson (1928–).



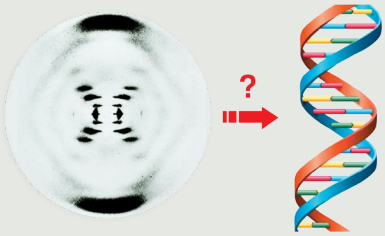
Francis Crick (1916–2004).

The following simulation helps us understand why Franklin's photo hinted at the double helical structure of DNA.

1 Use the transparency printed with patterns *A*, *B and C*.

2 Direct a laser beam on *A*, *B* and *C* in turn. Observe the patterns obtained behind the transparency.

When the laser beam is directed through *C*, can you see a similar diffraction pattern as the photo taken by Franklin?

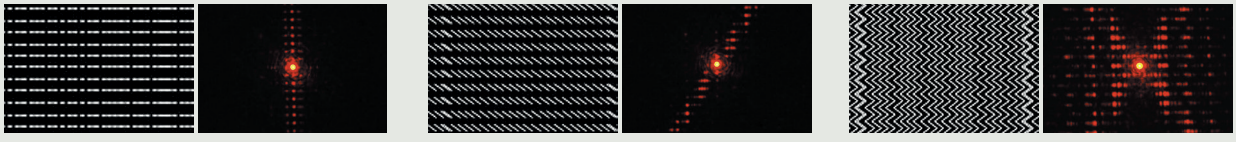


The X-ray diffraction photo (left) taken by Franklin hinted at the double helical structure (right) of DNA.

Atoms in DNA line up regularly, so DNA acts like an array of slits. The wavelength of X-rays is about the width of several atoms. Therefore, when X-rays pass through DNA, a diffraction pattern is obtained.

In the simulation, *A* acts as slits producing a vertical line of bright dots (diffraction pattern). *B* acts as slits at 45° to the horizontal. *C* acts as the double helix of DNA structure with pattern like two sets of intertwining slits at 45°, which form the bright dots in a cross shape.

pattern A *pattern B* *pattern C*

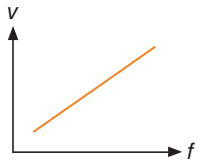


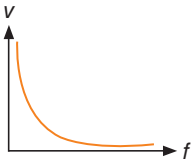
Patterns of slits and the diffraction patterns obtained.

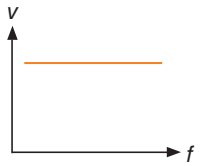
Rosalind Franklin 富蘭克林 James Watson 沃森 Francis Crick 克里克
Maurice Wilkins 威爾金斯

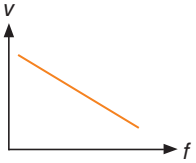
Exercise 6.4

★**1** Which of the following graphs shows the relation between the speed v and the frequency f of EM waves in a vacuum?

A 

B 

C 

D 

★**2** Which of the following correctly describe(s) the difference between infra-red radiation and ultra-violet radiation?

(1) They are different in colour.

(2) Ultra-violet radiation is a component of sunlight, but infra-red is not.

(3) They have different wavelengths.

A (2) only B (3) only

C (1) and (3) only D (1), (2) and (3)

★**3** When microwaves travel from air into water, which of the following properties of the waves will change?

(1) Speed (2) Frequency

(3) Wavelength

A (1) only B (2) only

C (1) and (3) only D (1), (2) and (3)

★**4** A new planet GJ 357 d, a world that could probably support life, was discovered by NASA's Transiting Exoplanet Survey Satellite (TESS) (Fig a). Light from GJ 357 d takes about 31 years to reach us.



Fig a

(a) Estimate the distance between GJ 357 d and the Earth.

(b) Suggest an electromagnetic wave which is suitable for TESS to communicate with the control centre on Earth. Give a reason for your answer.

★**5** In radio communications, radio waves of different frequencies travel a great distance in different ways. Low frequency radio waves travel along the Earth's surface while high frequency radio waves are reflected by a layer called the ionosphere in the atmosphere.

(a) Why are high frequency radio waves not suitable to travel a long distance along the Earth's surface?

(b) Suppose some high frequency radio waves from the transmitter travel 600 km to reach the ionosphere. The reflected waves then travel about the same distance to the receiver. How long would the transmission take?

★**6** A transmitter emits microwaves towards a convex paraffin lens (Fig b). The loudspeaker will produce a louder sound if the point detector detects a stronger microwave signal. When the detector is moved to different positions behind the lens, sounds of different loudness are produced.

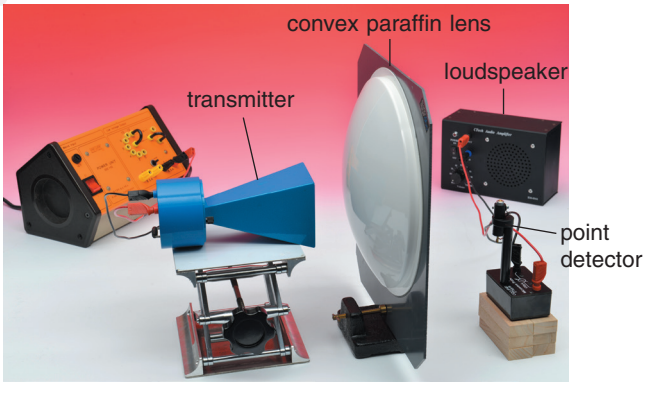


Fig b

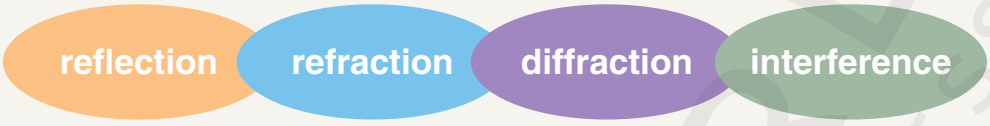
(a) What property of EM waves is demonstrated in the experiment?

(b) At a particular position, the loudest sound is produced. Explain briefly.

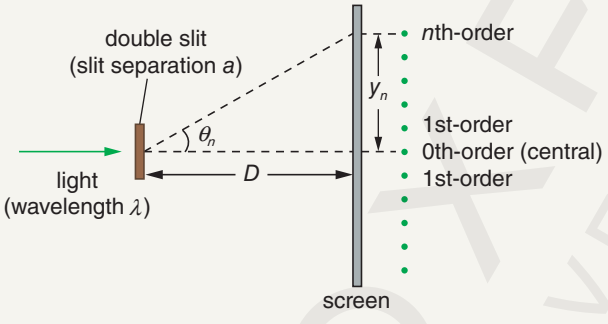
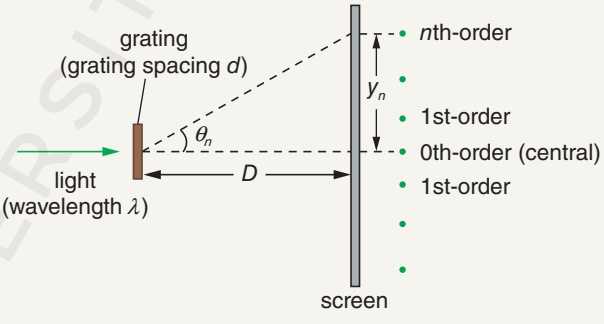
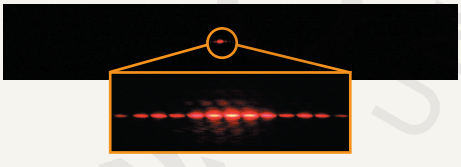
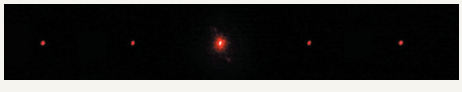
Key concepts 6

1 Wave nature of light

Light is an electromagnetic wave and exhibits:



2 Double slit and diffraction grating

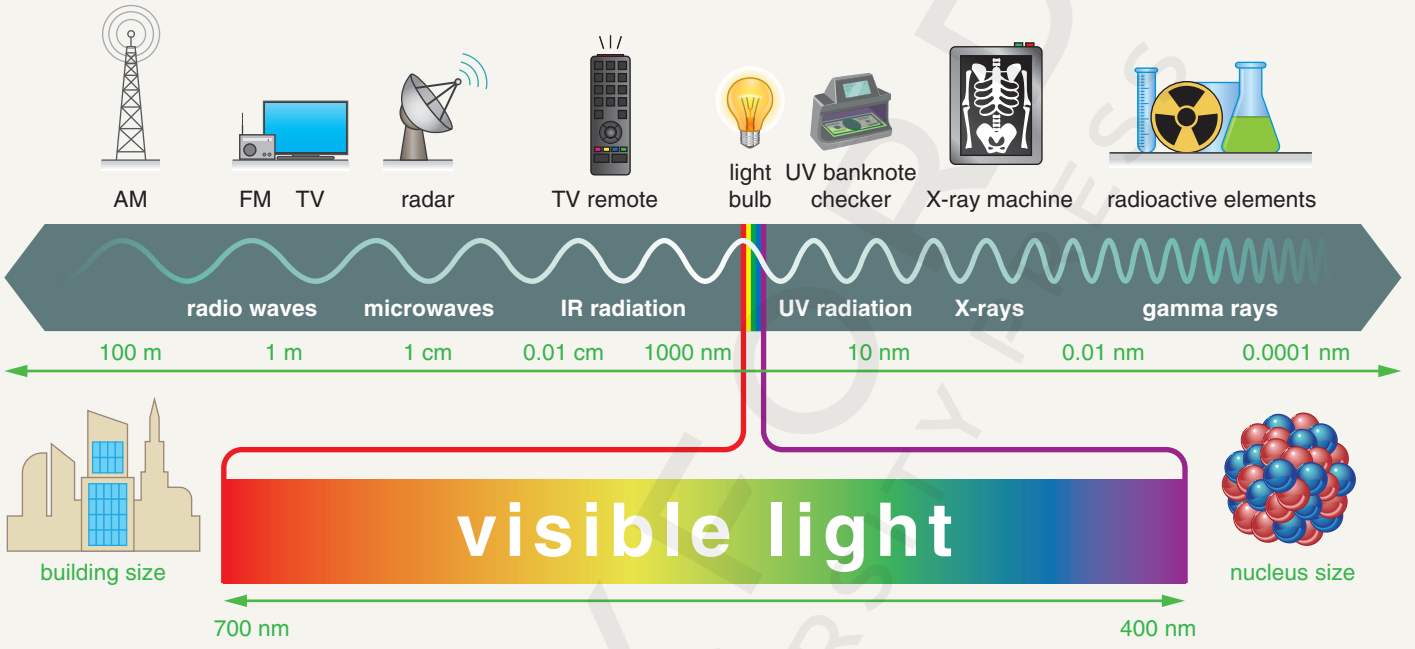
Double slit	Grating
	
 Pattern is dimmer and narrower.	 Pattern is brighter and wider.
Ext For $D >> a$, $a \sin \theta_n = n\lambda$	For $D >> d$, $d \sin \theta_n = n\lambda$
Ext For $D >> a$ and very small θ_n, Δy is constant. $\Delta y = \frac{\lambda D}{a}$	Δy is larger for fringes further from the centre.
Ext $y_n = n\Delta y$	$y_n = D \tan \theta_n$

The highest order of bright fringe formed by a grating is equal to the largest integer n satisfying

$$n \leq \frac{d}{\lambda}$$

3 Electromagnetic spectrum

- All EM waves are transverse waves which travel at 3×10^8 m s^{-1} in a vacuum.
- All EM waves obey $v = f\lambda$.
- All EM waves exhibit reflection, refraction, diffraction and interference.



EM waves	Wavelength in air / m
radio waves	$> 10^{-1}$
microwaves	$10^{-3} - 10^{-1}$
infra-red radiation	$7 \times 10^{-7} - 10^{-3}$
visible light	$4 \times 10^{-7} - 7 \times 10^{-7}$
ultra-violet radiation	$10^{-8} - 4 \times 10^{-7}$
X-rays	$10^{-11} - 10^{-8}$
gamma rays	$< 10^{-11}$

Key terms

1 diffraction grating 衍射光柵 p.116
2 double slit 雙縫 p.104
3 electromagnetic spectrum 電磁光譜 p.130
4 electromagnetic wave 電磁波 p.105
5 fringe 條紋 p.104
6 gamma ray 伽瑪射線 p.130
7 infra-red radiation 紅外輻射 p.130
8 microwave 微波 p.130
9 plane transmission grating 平面透射光柵 p.116
10 radio wave 無線電波 p.130
11 ultra-violet radiation 紫外輻射 p.130
12 X-ray X 射線 p.130
13 Young's double slit experiment 楊氏雙縫實驗 p.108

Revision 6

Multiple-choice questions

Take speed of light in a vacuum $c = 3.00 \times 10^8$ m s^{-1}.

1 Ext 6.2 In Young's double slit experiment, light of wavelength 570 nm is used. The distance between the two slits is 0.2 mm. On the screen, the separation between adjacent fringes is 1.7 mm. Which of the following gives the best estimation of the distance between the screen and the double slit?

A 0.50 m B 0.55 m
C 0.60 m D 0.65 m

***2** 6.3 Monochromatic light passes through a diffraction grating and an interference pattern is formed on a screen. Which of the following statements is/are correct?

(1) The bright fringes are evenly spaced.
(2) It is symmetric about the central bright fringe.
(3) Adjacent fringes may overlap.

A (2) only B (1) and (2) only
C (1) and (3) only D (1), (2) and (3)

***3** Ext 6.3 An interference pattern is formed by a diffraction grating. The angular position of the 1st-order bright fringe is 18°. What is the anglular position of the 2nd-order bright fringe?

A 9° B 33°
C 36° D 38°

***4** Misc Which of the following statements about the bright fringes formed by a double slit and a plane transmission grating is/are correct?

(1) The fringes formed by a grating are more widely spaced.
(2) The fringes formed by a double slit are brighter.
(3) The fringes formed by both apparatus are evenly spaced.

A (1) only B (2) only
C (1) and (2) only D (1) and (3) only

***5** Misc X and Y are visible lights of different colours. When they pass through a single slit, the diffraction pattern of X spreads out more than that of Y. This implies that

A X travels slower than Y.
B X travels faster than Y.
C the frequency of X is higher than that of Y.
D the frequency of X is lower than that of Y.

***6** 6.4 Which of the following statements about electromagnetic waves is/are correct?

(1) They all have very short wavelengths.
(2) They may travel at different speeds in different media.
(3) They carry energy.

A (2) only B (3) only
C (2) and (3) only D (1), (2) and (3)

***7** Misc We cannot observe the interference of light from two lamps. Which of the following is/are the possible reason(s) for this?

(1) The wavelength of visible light is very short.
(2) The speed of light is very high.
(3) Lights from the lamps are incoherent.

A (1) only B (1) and (3) only
C (2) and (3) only D (1), (2) and (3)

***8** Ext 6.3 A laser beam is incident on a plane transmission grating that has 5300 lines per centimetre. The zeroth-order and first-order bright fringes are separated by 0.48 m on a wall 1.7 m away from the grating. What is the distance between the 1st-order and 2nd-order bright fringes?

A 0.48 m B 0.62 m
C 0.96 m D 1.10 m

***9** 6.4 Infra-red radiation undergoes diffraction after passing through a slit. Which of the following changes can increase the amount of diffraction?

(1) Increase the width of the slit.
(2) Increase the wavelength of the infra-red radiation.
(3) Increase the amplitude of the infra-red radiation.

A (1) only B (2) only
C (1) and (3) only D (2) and (3) only

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***10** Ext 6.3 A green laser beam of wavelength 532 nm is directed towards a diffraction grating (Fig a). The screen is 0.8 m from the grating. The distance between the central and first-order bright fringes on the screen is 8.5 cm.

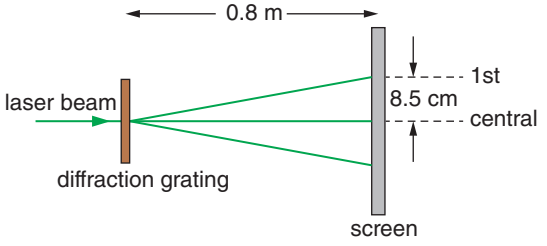


Fig a

Estimate the number of lines per mm of the diffraction grating.

A 5×10^{-6} B 5×10^{-3}

C 2.0×10^{2} D 2.0×10^{5}

****11** Ext 6.3 A beam of white light is directed perpendicularly onto a plane transmission grating (4000 lines per cm) and a pattern is formed on a screen 0.8 m from the grating. Estimate the width of the first-order spectrum in the pattern.

A 10 cm B 13 cm

C 16 cm D 23 cm

Refer p.122

****12** Ext 6.3 X and *Y* are two diffraction gratings, with the grating spacing of *X* being 3 times that of *Y*. When monochromatic light is incident on *X*, a fringe pattern is formed and marked on a screen. After *X* is replaced with *Y*, another fringe pattern is formed on the screen. Which of the following statements is/are correct?

(1) For both *X* and *Y*, the angles of diffraction of light are directly proportional to the wavelength of light.

(2) The 3rd-order maximum produced by *X* coincides with the 1st-order maximum produced by *Y*.

(3) The angle of diffraction of the 2nd-order maximum produced by *Y* is 3 times that produced by *X*.

A (1) only B (2) only

C (1) and (2) only D (2) and (3) only

Refer p.120

Q13 Exam Report: Less than half were able to identify the two correct spectra formed from white light in the diagrams.

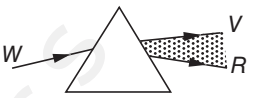
13 Misc **HKDSE 2015 Paper 1A Q17**

Which diagrams below correctly show the spectra formed from white light by a glass prism and a diffraction grating respectively? It is known that red light travels faster than violet light in glass. (*R* = red, *V* = violet, *W* = white)

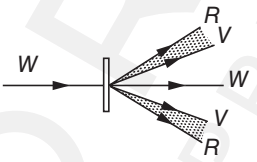
(1)



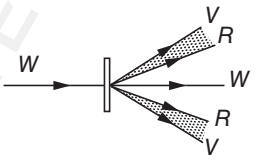
(2)



(3)



(4)



A (1) and (3) only (45%)

B (1) and (4) only

C (2) and (3) only

D (2) and (4) only

14 Ext 6.3 **HKDSE 2017 Paper 1A Q20**

Figure b shows some of the bright fringes formed when monochromatic light passes through a diffraction grating.

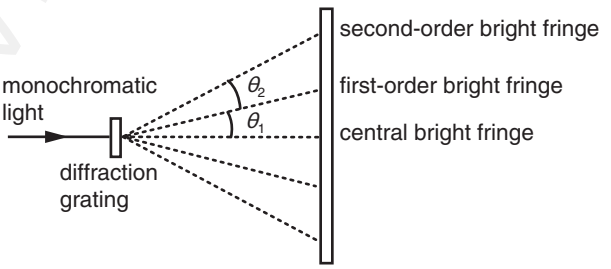


Fig b

Which of the following is/are correct?

(1) $\theta_1 = \theta_2$

(2) The maximum order of bright fringe is 4 if $\theta_1 = 20°$.

(3) θ_1 will decrease if the experiment is performed in water but not in air.

A (1) only

B (3) only (63%)

C (1) and (2) only

D (2) and (3) only

15 **HKDSE 2020 Paper 1A Q15**

Ext 6.2

Figure c shows a typical set-up of Young's double slit experiment.

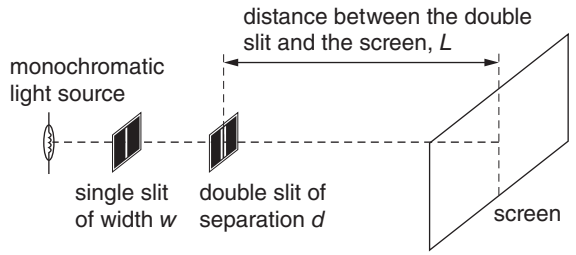


Fig c

Which combination below is the best setting for displaying an observable fringe pattern on the screen?

	w	*d*	*L*
A	0.1 mm	1 mm	10 m
B	0.1 mm	1 mm	1 mm
C	1 mm	0.1 mm	1 mm (52%)
D	1 mm	0.1 mm	0.1 mm

16 **HKDSE 2021 Paper 1A Q19**

6.4

Typical wavelength of *X*-ray and microwaves are in the ratio 10^n : 1. The value of *n* could be

A −10. (56%) B −4.

C +4. D +10.

Conventional questions

★**17** 6.4

(a) Which type of electromagnetic wave has the longest wavelength and which has the shortest? (2 marks)

(b) The weather radar at Tai Mo Shan (Fig d) sends a pulse of microwave at a frequency of 1 GHz. The reflected pulse is received after 1.8×10^{-5} s. Given: 1 GHz = 10^9 Hz.



Fig d

(i) What is the wavelength of the microwave? (2 marks)

(ii) What is the distance between the radar and the object that reflects the microwave pulse? (2 marks)

★**18** Misc

Red light and blue light are directed in turn to device *X* (Fig e). Figures f and g show the interference patterns formed on the screen using the same scale.

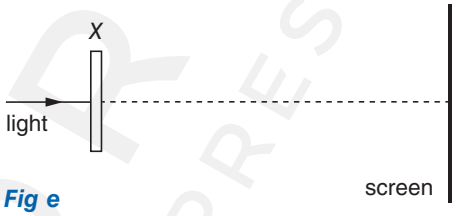


Fig e

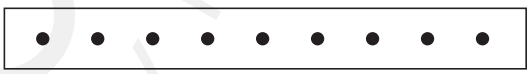


Fig f

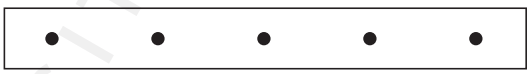


Fig g

(a) Is device *X* a double slit or a transmission grating? Explain. (2 marks)

(b) Which figure shows the interference pattern of the red light? Explain. (2 marks)

(c) Mary claims that more fringes can be captured on the screen if the screen is moved towards *X*. Explain whether she is correct. (2 marks)

★**19** 6.2

In a double slit experiment, red light is used and a pattern consisting of bright fringes is projected on a screen.

(a) Describe the change in the fringe separation if the following modifications are made independently.

(i) Increasing the slit separation (1 mark)

(ii) Moving the screen further from the double slit (1 mark)

(iii) Using blue light instead of red light (1 mark)

Ext (b) Explain your answers in (a) by considering a suitable formula. (4 marks)

★20 Ext 6.2 You are given a double slit, a laser pointer, a screen and a metre rule (Fig h).

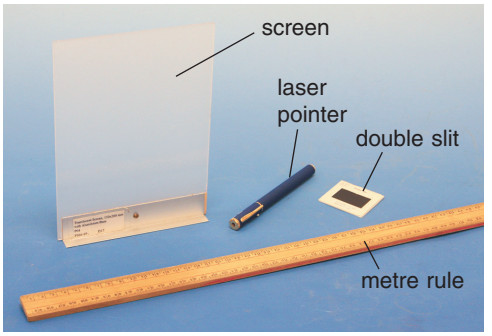


Fig h

(a) Describe an experiment to estimate the wavelength of the light emitted from the laser pointer. Write the equations involved in calculating the wavelength. (5 marks)

(b) State one safety precaution for the experiment in (a). (1 mark)

(c) Explain why replacing the double slit with a diffraction grating can improve the accuracy of the result of the experiment in (a). (2 marks)

★21 Ext 6.2 Monochromatic light is directed onto a double slit with slit separation 6×10^{-4} m. Bright fringes with a fringe separation of 0.7 mm are formed on a screen 1.2 m behind the double slit.

(a) Find the wavelength of light. (2 marks)

(b) The screen is then moved to a position which is 6 m behind the double slit. Find the fringe separation. (1 mark)

(c) The double slit is replaced by a diffraction grating. Write down two changes of the bright fringes. (2 marks)

★22 Ext 6.3 Light of wavelength 610 nm is incident on a plane transmission grating which has 4000 lines per cm.

(a) Find the grating spacing (in m). (1 mark)

(b) Find the angle of diffraction of the 2nd-order maximum. (2 marks)

(c) Find the highest order of maximum produced. (2 marks)

(d) By using the same grating, suggest one method to increase the number of maxima produced. (1 mark)

★23 Ext 6.3 A student directs orange light onto a plane transmission grating (3200 lines per cm) (Fig i). The pattern is projected on a screen which is 80 cm from the grating.

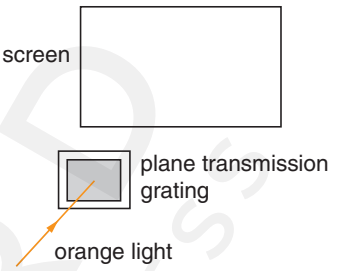


Fig i

(a) If the distance between the third-order maximum and the central maximum is 55 cm, what is the angular position of the third-order maximum? (1 mark)

(b) Hence, find the wavelength of the orange light. (2 marks)

(c) The student claims that the angular position of the first-order maximum is one third the angular position of the third-order maximum. Explain whether he is correct. (2 marks)

★24 Misc A beam of UV radiation of wavelength 100 nm enters medium X from the air (Fig j). The green lines show its wavefronts and the orange line shows its travelling direction. It travels at 2.16×10^8 m s^{-1} in X.

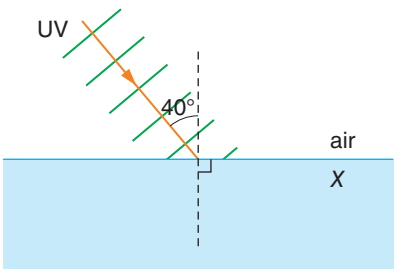


Fig j

(a) Find the wavelength λ of the UV radiation in X. (2 marks)

(b) Find the frequency f of the UV radiation in air and in X. (3 marks)

(c) Find the angle of refraction. (2 marks)

(d) Sketch the wavefronts of the UV radiation in X. (2 marks)

★25 Misc Read the following passage about **laser speed guns** and answer the questions that follow.

Police use laser speed guns to monitor the speed of moving cars (Fig k). The gun shoots a very short burst of infra-red laser and measures the round-trip time, i.e. the time it takes for the pulse to reach a car and get reflected back. The distance of the car from the speed gun can then be calculated instantly. After a short time, the gun shoots another pulse to obtain the new distance of the car. From the change in distance, the speed of the car can be calculated. One of the advantages of the laser speed gun is that it allows the police to target a specific vehicle. The movement of other vehicles does not affect the result.



Fig k

On a highway with a speed limit of 100 km h^{-1}, a policeman suspects that a car is speeding. He aims the laser speed gun at the car and gets two round-trip times of 300.00 ns and 300.24 ns respectively. The two measurements are taken 0.001 s apart.

(a) Find the distances of the car from the speed gun when the two measurements were taken. (3 marks)

(b) Hence find the speed of the car in km h^{-1}. Is the car speeding? (2 marks)

★26 Misc An aluminium sheet with two narrow slits, *X* and *Y*, is placed in front of a microwave transmitter. *X* and *Y* are equidistant from the transmitter (Fig l). *P* is a point on the other side of the aluminium sheet and is equidistant from *X* and *Y*. A receiver connected to a computer is moved along *PR*.

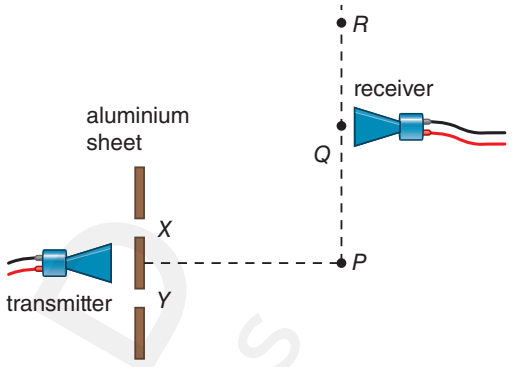


Fig l

The computer records the data and generates a graph showing the amplitude of the microwaves received along *PR* (Fig m).

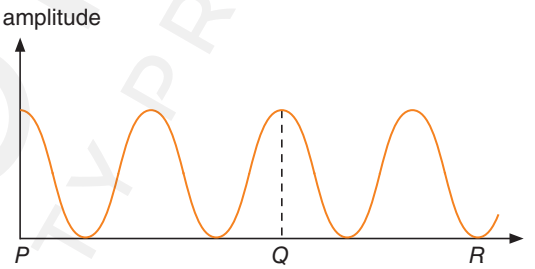


Fig m

(a) Explain the shape of the graph. (3 marks)

(b) If *XQ* = 21 cm and *YQ* = 25 cm, find the wavelength of the microwaves. (2 marks)

(c) The separation between the slits is then reduced such that a peak is still recorded at *Q*. Express the new slit separation in terms of the original slit separation *a*. (1 mark)

★27 Ext Misc A remote control is vertically above a red LED. A diffraction grating with 100 lines per mm is 1.1 m away from them. A camera is put behind the grating (Fig n). The wavelength of the light emitted by the LED is 630 nm.

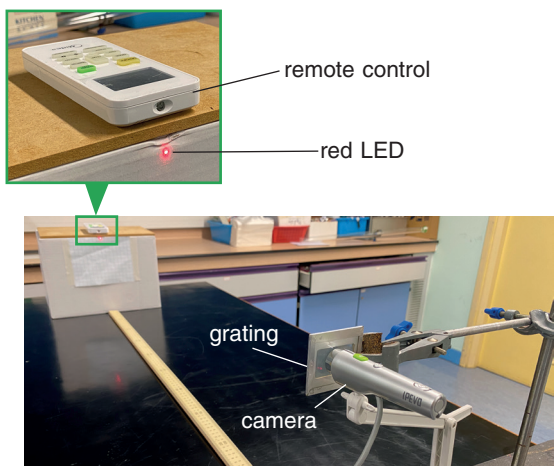


Fig n

When a button on the remote control is pressed, an IR signal is emitted. Figure o shows the photo taken by the camera.

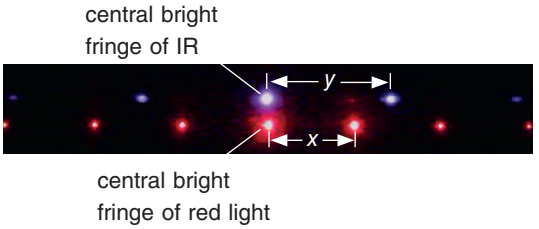


Fig o

(a) Find the angular position of the first-order bright fringe of red light. (2 marks)

(b) Find the distance x. (1 mark)

(c) The ratio $x : y$ is equal to 0.698 : 1. Estimate the wavelength of the IR signal. (3 marks)

★★28 Ext 6.3 Light of wavelength 4.8×10^{-7} m is incident on a diffraction grating with 3500 lines per cm. The bright fringes formed on the screen behind the grating are marked.

(a) Find the frequency of the incident light. (2 marks)

(b) Find the highest order of fringe produced. (2 marks)

(c) The light source is then replaced by one emitting monochromatic light of another wavelength. In the newly formed pattern, a bright fringe is observed at nearly the same position as the 4th-order fringe in the previous pattern.

(i) Find the angular position of that fringe. (2 marks)

(ii) Given that the wavelength of visible light ranges from 400 nm to 700 nm, estimate the wavelength of the light. (2 marks)

Refer p.122

★★29 6.2 Dennis directs yellow light towards a double slit and a pattern is formed on the screen behind the double slit.

(a) State one precaution that should be taken in this experiment to produce an observable pattern. (1 mark)

(b) What would happen to the pattern if blue light is used instead of the yellow light? (1 mark)

(c) Ext Explain your answer in (c) by considering a suitable formula. (2 marks)

(d) Ext The wavelength of the incident yellow light is 590 nm and the slit separation of the double slit is 0.15 mm. If observations can be carried out up to an angle of 2° on both sides of the central maximum, how many bright fringes are observed? (3 marks)

Refer p.114

See p.148 for the Exam Report.

30 HKDSE 2013 Paper 1B Q7 Ext Misc

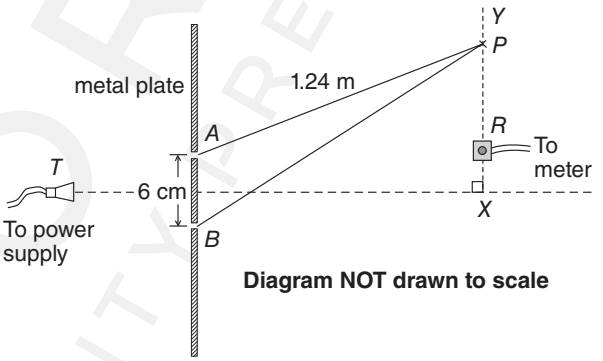


Fig p

Figure p shows a set-up for the study of interference of microwaves. Microwaves of wavelength 2 cm emitted from a transmitter T pass through two slits A and B formed by metal plates. The slits are separated by 6 cm as shown. A probe R connecting to a meter is moved from X to Y to detect the intensity of microwaves received. Transmitter T and point X are equidistant from A and B.

(a) Calculate the frequency of the microwaves. (2 marks)

(b) (i) The meter shows alternate maxima and minima when R moves along XY. Explain. (2 marks)

(ii) The second minimum is found at position P where AP = 1.24 m. Find BP. (2 marks)

(iii) When R is moved along XY from X towards Y and beyond, explain whether or not it is possible to detect more than three maxima. (2 marks)

(c) Microwaves can be used in radar. Why are radio waves of lower frequencies not suitable for use in radar? (2 marks)

See p.148 for the Exam Report.

31 HKDSE 2014 Paper 1B Q7

Ext 6.3

Figure q shows an experimental set-up to determine the wavelength of monochromatic light emitted from the vertical narrow slit of a discharge lamp. *A*, *B* are two mutually perpendicular metre rules on the bench with rule *A* pointing towards the lamp. A diffraction grating with vertical lines is placed at one end of rule *A*. A vertical mounted pin *P* is moved along rule *B* until the pin is in line with the diffracted image of the second-order to the observer. The corresponding distance *x* is measured for finding the diffraction angle θ.

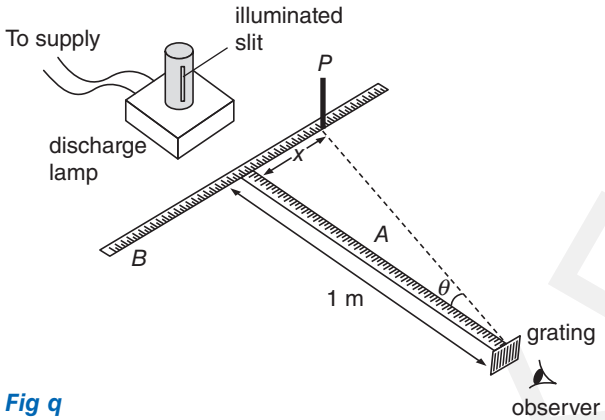


Fig q

The grating has 300 lines per mm and *x* is found to be 0.38 m for the **second-order** image.

(a) (i) Calculate the diffraction angle θ. (1 mark)

(ii) Hence find the wavelength of the light from the lamp. (3 marks)

(iii) Give **one** advantage of measuring the position of the second-order image instead of the first-order one. (1 mark)

(b) In the experiment, the illuminated slit may not be well aligned along metre rule *A*. Suggest one way to reduce this error. (2 marks)

See p.148 for the Exam Report.

32 HKDSE 2015 Paper 1B Q7

Ext 6.2

(a) A laser light beam of wavelength 650 nm is incident normally on a pair of slits separated by a = 0.325 mm. Interference pattern is observed on a screen at a distance D = 3.0 m from the slits as shown in Figure r. What is the separation between adjacent first- and second-order bright fringes? (2 marks)

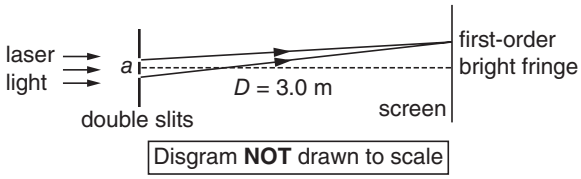


Fig r

(b) Figure s shows a set-up with two small LEDs separated by 6 mm and both LEDs emit light of wavelength 650 nm. State and explain what you would expect to see on the screen. (2 marks)

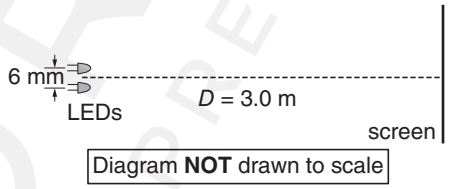


Fig s

Figure t shows circular water waves in a ripple tank. The two point sources S_1 and S_2, separated by 40 mm, are driven by the same vibrator. The solid lines represent the wave crests from S_1 and the dotted lines represent the wave crests from S_2. The wavelength of the waves is 10 mm.

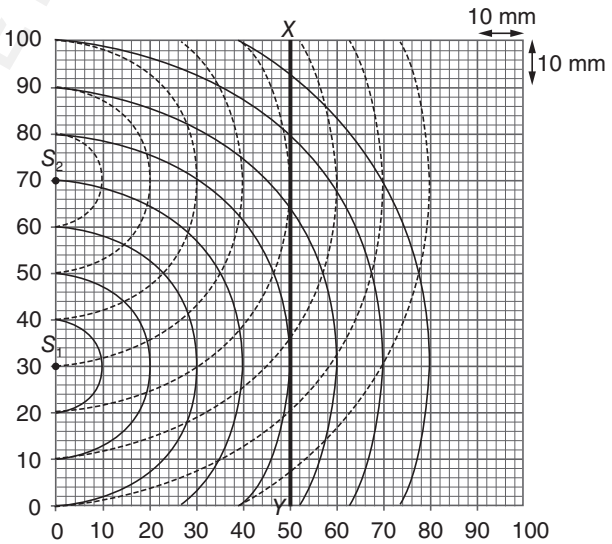


Fig t

(c) Sketch on Figure t two lines to indicate all points *P* with path difference $PS_1 - PS_2$ equals to 10 mm (L_1) and 20 mm (L_2). State the kind of interference that occurs at these points *P*. (3 marks)

(d) (i) If the interference pattern is observed along line XY at 50 mm from the sources as shown, **measure** the separation between adjacent first- and second-order maxima Δy. (1 mark)

Separation Δy = _____

(ii) However, using the calculation method in (a) would obtain 12.5 mm for this separation. Why does this calculated value differ with the measurement in (d)(i)? (2 marks)

33 HKDSE 2016 Paper 1B Q6

Ext Misc

(a) A laser beam is directed perpendicularly towards a double slit of separation a = 0.3 mm. The pattern of bright spots projected on a screen 1.8 m away from the slits is shown in Figure u.

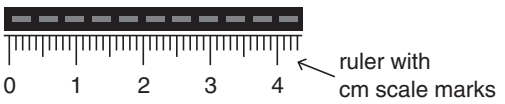


Fig u

(i) Find the wavelength of the laser beam. (3 marks)

(ii) Explain why the slit width has to be very narrow in order for the above pattern to be observed. (2 marks)

(b) The double slit is now replaced by a diffraction grating of 500 lines per mm.

(i) Find the separation between the central bright spot and first-order bright spot of the pattern on the screen for the same experimental settings. (3 marks)

(ii) Sketch the pattern, up to the second-order, that you would expect to see on the screen when using this diffraction grating. A first-order bright spot has already been drawn for you. (2 marks)

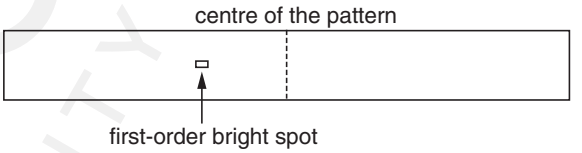


Fig v

Q30 Exam Report: Most candidates answered part (a) correctly although a few of them forgot to convert the unit of wavelength to metres. Some candidates were not able to demonstrate an understanding that microwaves travel at the speed of light and a few even mistook it as the speed of sound (340 m s^{-1}). In (b)(i), candidates showed an understanding that the alternate maxima and minima were due to constructive and destructive interference respectively, but very few stated the reason for this was due to variation of path difference along XY. In (b)(ii), some candidates failed to realize that the path difference concerned was 1.5λ or mistook it as $AP - BP$. Candidates' performance in (b)(iii) was poor. Many did not count the zeroth order maximum or incorrectly stated that the order corresponding to θ = 90° could still be observed. The equation $d \sin\theta = n\lambda$ was also incorrectly applied as the slit separation was not negligible in such a situation. In (c), most candidates knew that the diffraction effect of radio waves was more significant compared to microwaves but very few mentioned how the reflection of waves from small obstacles would be affected as a result.

Q31 Exam report: Candidates performed well in parts (a)(i) and (ii) involving the calculations of diffraction grating. Many did not have a clear concept of percentage error in x or θ and therefore failed to score marks in (a)(iii). It seems that most candidates did not understand what part (b) asked for and gave answers like using a grating with more lines per mm.

Q32 Exam report: Part (a) was well answered. Some candidates made mistakes in converting units when calculating the fringe separation. In (b), few candidates realized that interference did occur but was unobservable as the sources were incoherent. Many thought that an interference pattern with smaller fringe separation formed would occur as the source separation was greater. Some confused phase and coherence, and stated 'the interference was unobservable because they were out of phase'. Quite a number of them sketched straight lines instead of curves for the antinodal lines in (c). In (d)(i), more than half of the candidates incorrectly read the value of y even though correct antinodal fines were drawn as they might have overlooked the scale (10 mm per division) given. Part (d)(ii) was poorly answered as most candidates did not know the geometrical constraints in estimating the fringe separation.

Q33 Exam report: Although most candidates applied the correct formula to find λ in (a)(i), many made mistakes in getting the fringe separation from the pattern given or in converting the units. In (a)(ii), quite a number of the candidates confused slit width with slit separation. Not many pointed out explicitly that the diffracted waves from the slits have to overlap for interference to occur. Some candidates wrongly applied $\Delta y = \frac{\lambda D}{a}$ to find the separation between bright spots in (b)(i). Candidates' performance in (b)(ii) was poor. The patterns drawn were either with uniform separation between bright spots, not symmetrical about the centre or without the central bright spot.

Know more

Deriving $\Delta y = \frac{\lambda D}{a}$ (p.112)

Refer to Figure 6.2i on p.112.

In $\triangle XYZ$, $\tan \theta_n = \frac{y_n}{D}$ ..... (1)

Since D is much greater than a, we can take that

$AZ \parallel XZ \parallel BZ$

In other words, both light rays are at an angle θ_n to XY (Fig a).

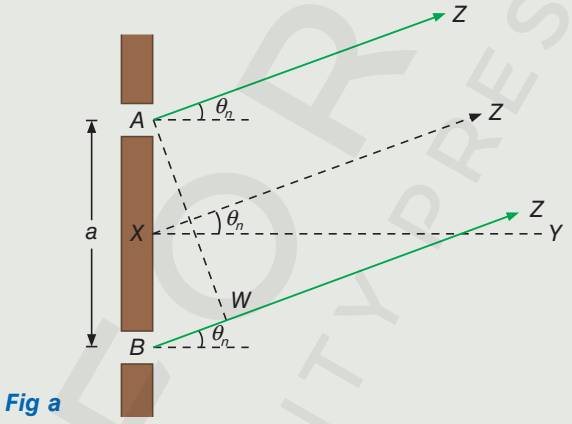


Fig a

In Figure a, W is a point on BZ such that $AZ = WZ$ and $\angle AWB$ is approximately 90°.

In $\triangle ABW$, $\angle BAW = 180° - \angle AWB - \angle ABW$ ($\angle$ sum of Δ)

$= 180° - 90° - \angle ABW$

$= 90° - \angle ABW$

$= \theta_n$

In addition, $BW = AB \sin \angle BAW$

$\therefore BW = a \sin \theta_n$ ..... (2)

Path difference of the two light rays at $Z = BW$

A bright fringe is formed when the two rays from the double slit undergo constructive interference. This happens when the path difference at that point is equal to a whole number of wavelengths. For the nth-order bright fringe, the path difference is $n\lambda$.

From (2), we have:

$a \sin \theta_n = n\lambda$ ..... (3)

In practice, the size of the whole interference pattern observed is very small when compared to the distance D. As a result, the value of θ_n is small and allows us to use the small angle approximation, which is

$\sin \theta_n \approx \tan \theta_n$ ..... (4)

Substitute (1) and (4) into (3), we have

$$a \times \frac{y_n}{D} = n\lambda$$

$$y_n = \frac{n\lambda D}{a}$$

Similarly, for the $(n + 1)$th-order bridge fringe, we have

$$y_{n+1} = \frac{(n + 1)\lambda D}{a}$$

Therefore, the fringe separation is

$$\Delta y = y_{n+1} - y_n = \frac{(n + 1)\lambda D}{a} - \frac{n\lambda D}{a} = \frac{\lambda D}{a}$$

Know more

Derivation of $d \sin\theta = n\lambda$ (p.120)

Refer to Figure a. Consider all light rays (one from each slit) arriving at Z. Since distance D is much greater than grating spacing d, we can consider all the rays to be parallel to each other and at angle θ_n to the central axis (Fig b).

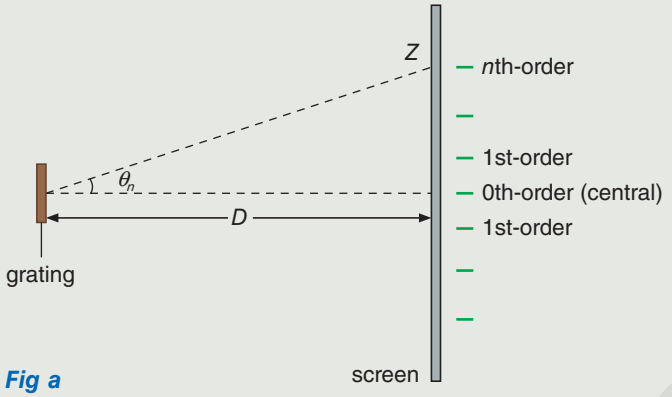


Fig a

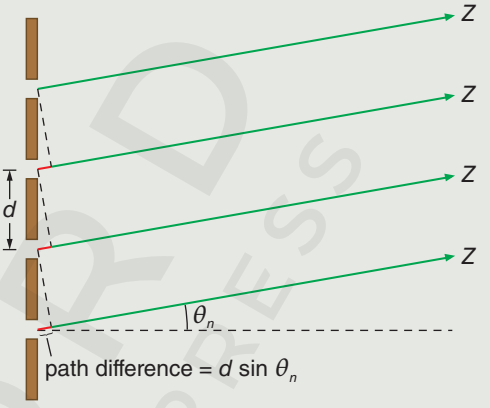


Fig b

Then, the path difference Δ of two adjacent rays is given by

$$\Delta = d \sin\theta_n \quad \text{..... (1)}$$

A bright fringe is formed when all the rays from the grating undergo constructive interference. This happens when the path difference of two adjacent rays is equal to a whole number of wavelengths. For the nth-order bright fringe, the path difference is $n\lambda$.

From (1), we have:

$$d \sin\theta_n = n\lambda$$

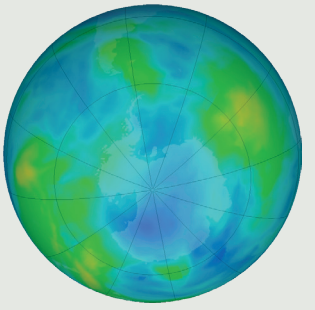
Unlike Young's double slit experiment, θ_n is usually not small and so we cannot apply the small angle approximation.

STSE

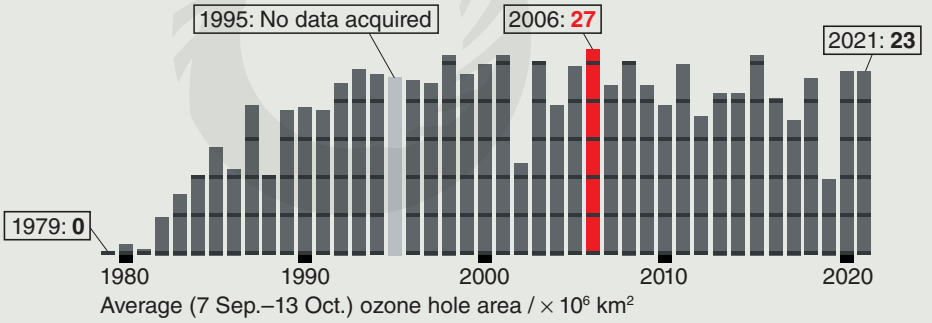
UV threat: Depletion of the ozone layer (p.132)

Ozone (O_3) is a colourless gas. Most of the ozone in the atmosphere is found in a layer between 15 km and 35 km above the Earth's surface. The ozone layer acts as a protective filter which blocks most of the harmful UV radiation from the Sun.

Scientists first announced the ozone hole over the South Pole in 1985. This was caused by the excessive release of certain chemicals produced through human activities, especially chlorofluorocarbons (CFCs). The production and use of CFC products was sharply limited as a result. In 2016, it was reported that the layer showed a gradual trend towards recovery.



Ozone hole (blue region) over the South Pole on 26 May 2023.



Average ozone hole area from 7 September to 13 October for each year (Data obtained from NASA website in Oct 2022).

Know more

Colours produced by interference (p.123)

Butterfly

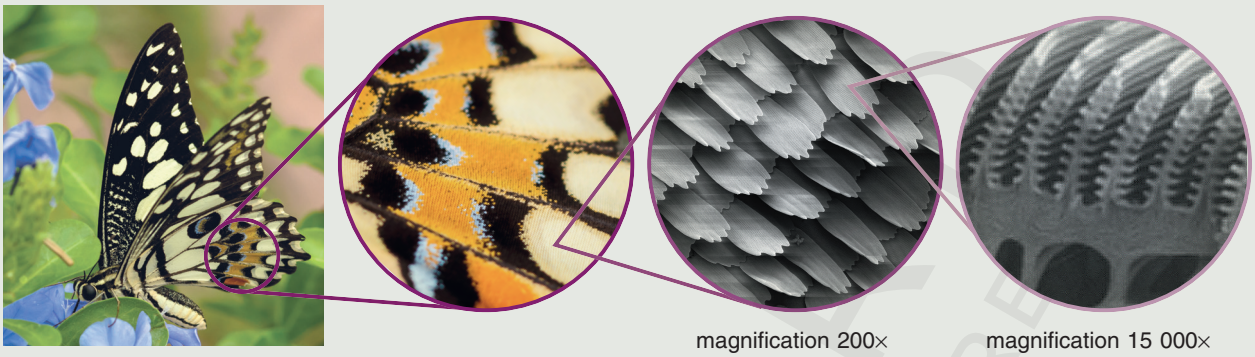


Fig a

The wings of butterflies consist of a colourless membrane covered by scales. Each scale contains multilayer fine structures (Fig a).

Light shone on the wings is reflected by each layer. For light of a certain colour (i.e. wavelength), the reflected light waves from every layer undergo constructive interference (path difference of the light waves = $n\lambda$) (Fig b), making this colour much brighter. At different parts of the wing, the thicknesses of the layers are different and therefore different colours are shown. This explains why the wings of butterflies appear so colourful.

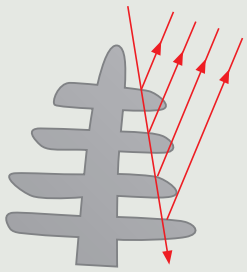


Fig b

DVD grating

A DVD looks very colourful though it is plain in colour (Fig c). The colours change when you look at it at different angles. The following explains this effect.

A DVD records data on circular tracks. The tracks are separated by a minimum distance of 740 nm, i.e. about 1350 tracks per mm. When light falls on it, it acts like a transmission grating, except that light is reflected on the surface instead of passing through it. Different colours undergo constructive interference at different angles. As a result, we see different colours at different parts (i.e. different angles to the eye) on the disc.



Fig c

Seeing and hearing IR (p.131)

You can see the IR emitted by a remote control through the cameras of some smartphones (Fig d).

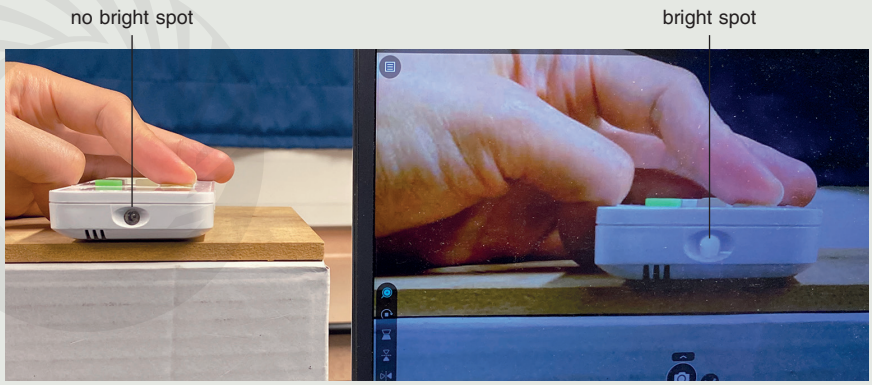


Fig d

Different remote controls emit IR of different frequencies. By connecting an IR receiver to a loudspeaker and using different remote controls to send signal to the receiver, different sound can be heard!

Video

→ This video shows the different sound produced by different remote controls.

Know more

Grating in daily life (p.123)

When you look at distant street lights through a gauze curtain, the lights appears as small crosses rather than dots (Fig a). Do you know why?

(i) *(ii)*



Fig a *View (i) without (ii) with a gauze curtain.*

When you look at a small light source through a coarse grating with vertical slits, a horizontal line of closely packed fringes can be seen (Fig b). If the grating is rotated by 90° so that the slits become horizontal, the fringes will be located along a vertical line (Fig c)

If the light source is observed through two mutually perpendicular coarse gratings, interference will occur in both the horizontal and vertical directions. A two-dimensional interference pattern will be seen (Fig d). If the pattern is observed from a distance, the individual fringes will become undistinguishable and a cross pattern will be produced.

A gauze curtain has a mesh-shaped structure which acts like two mutually perpendicular coarse gratings. Hence, a distant light appears as a cross when viewed through a gauze curtain.

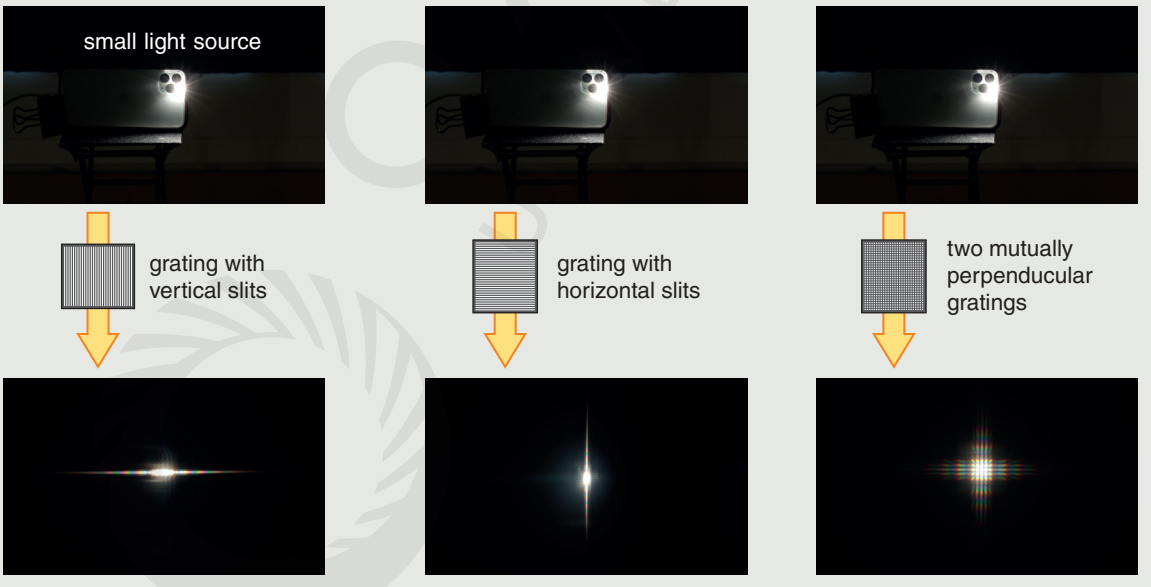


Fig b ***Fig c*** ***Fig d***